VPN
Virtual Private Network

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my laptop from Germany is connected to polito, and get the same access, but he is using a shared infrastructer, as if the laptop was connected into the private network of the university.

how can we make an host in the network as if it was in another place in the world? trough tunneling. and crypthography because we want to avoid that someone pretends to be us.

A Definition

Virtual Private Network

Connectivity realized on a shared infrastructure such that policies can be enforced as in a private network

when u join a vpn in another country u are threathed as if u were in another country.

as if it was part of another network somewhere else.

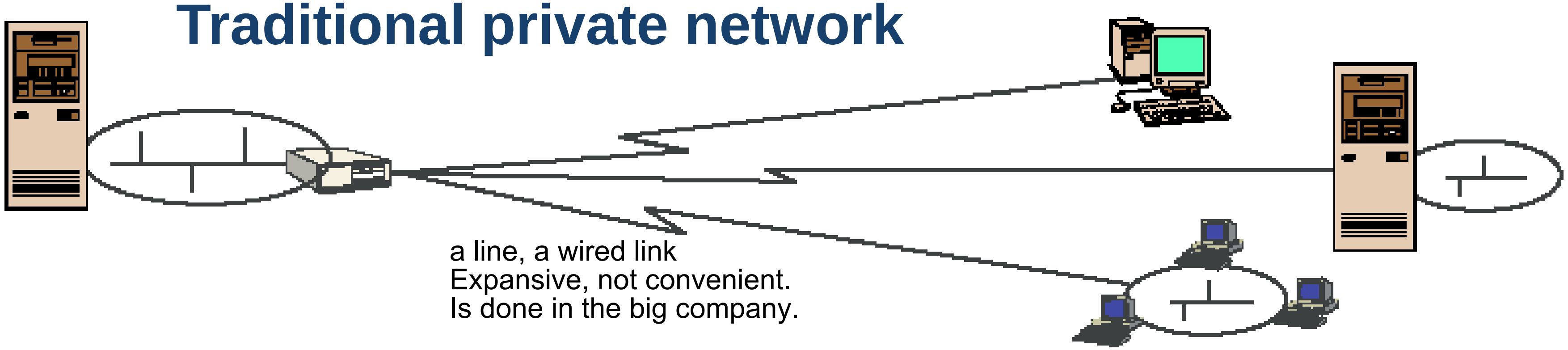
- **Shared infrastructure:**
  - **Private/public network**
    - **e.g., the one of an Internet Service Provider**
    - **IP**
    - **Frame Relay**
    - **ATM**
  - **The Internet**
- **Policies**
  - **Security, Quality of Service (QoS), reliability, addressing, etc.**

same access permission as if we were in polito, whereas we are in Germany

Secure communication

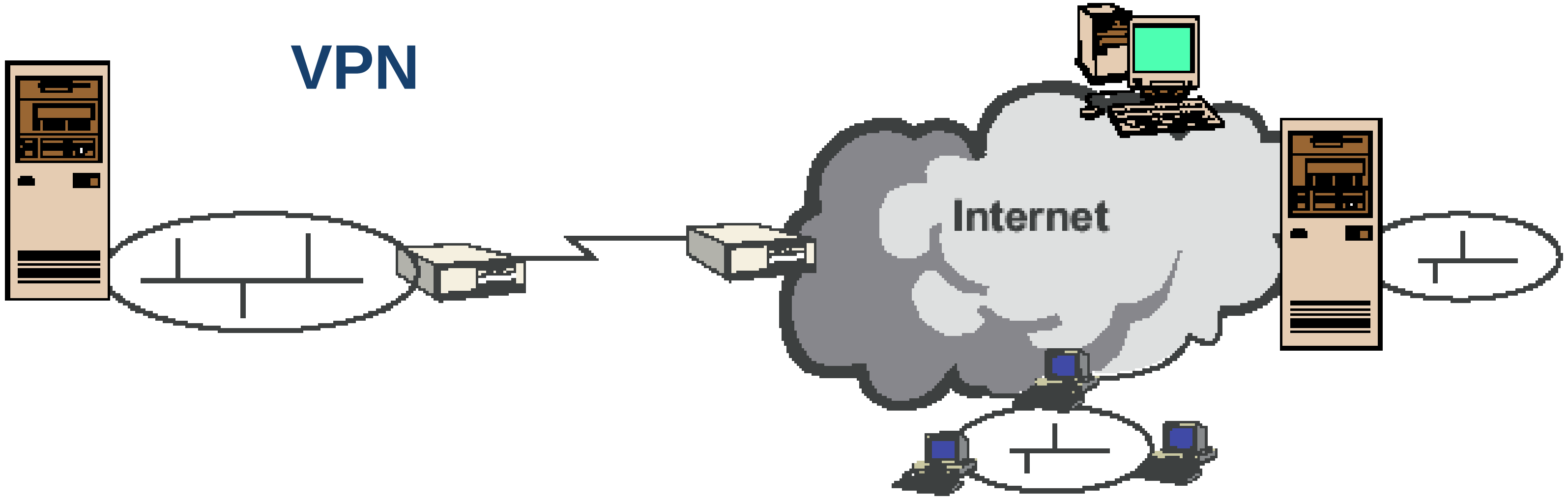
Sample Use Case

Traditional private network

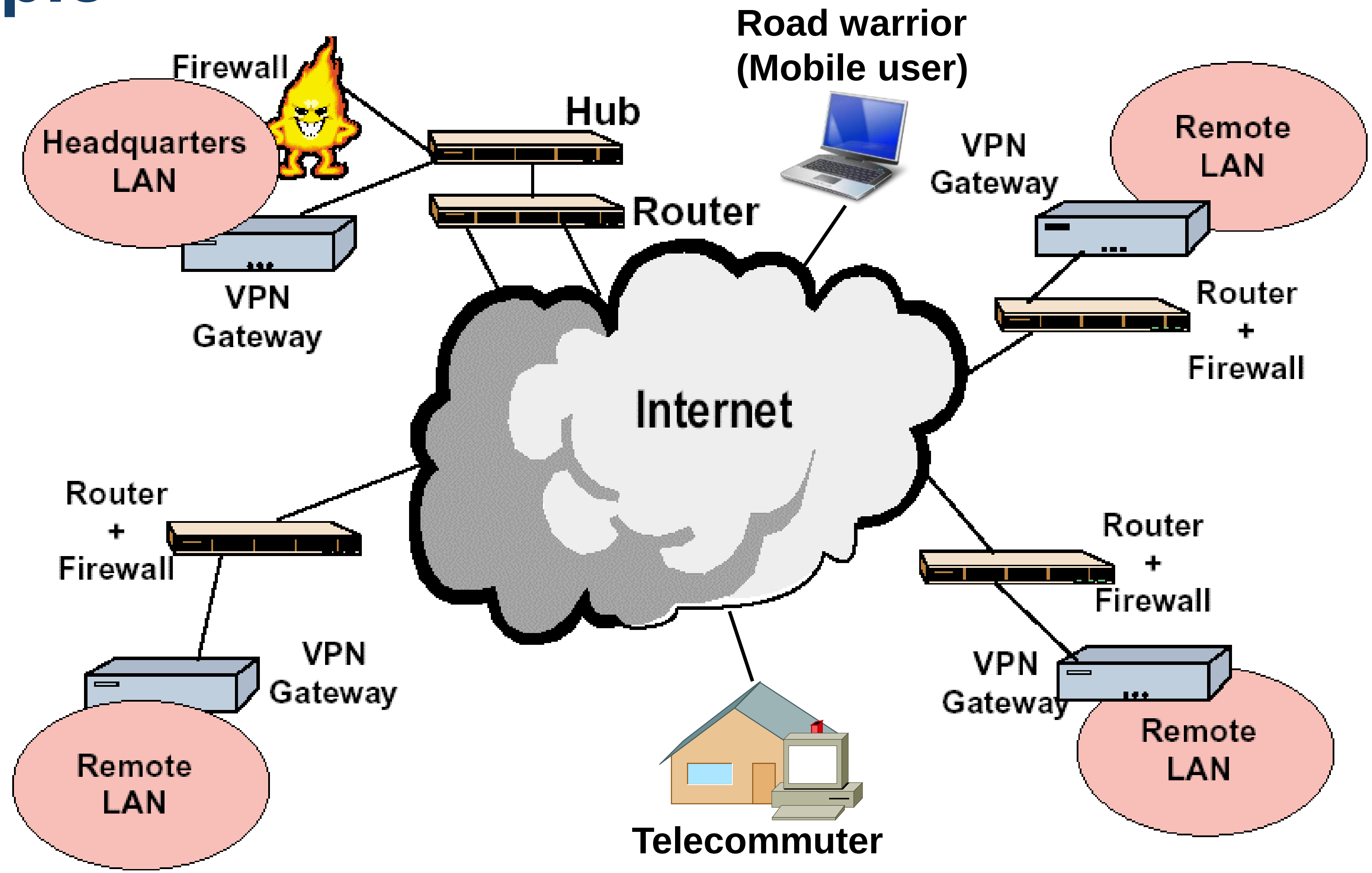


a line, a wired link
Expansive, not convenient.
Is done in the big company.

VPN



Example of VPN



Key Elements

- **Tunnel** put a packet into another packet, used only on that shared infrastructure.
  - **(Secure) encapsulation of corporate traffic while in transit on the shared network**
  - **Not present in some solutions**
- **VPN Gateway** usually terminate tunnels.
  - **Termination device on the corporate network**
  - **Might be a tunnel endpoint**

We will get back to these later on

Motivations

Why VPN?

because they are cheaper and not so expansive like wired networks.

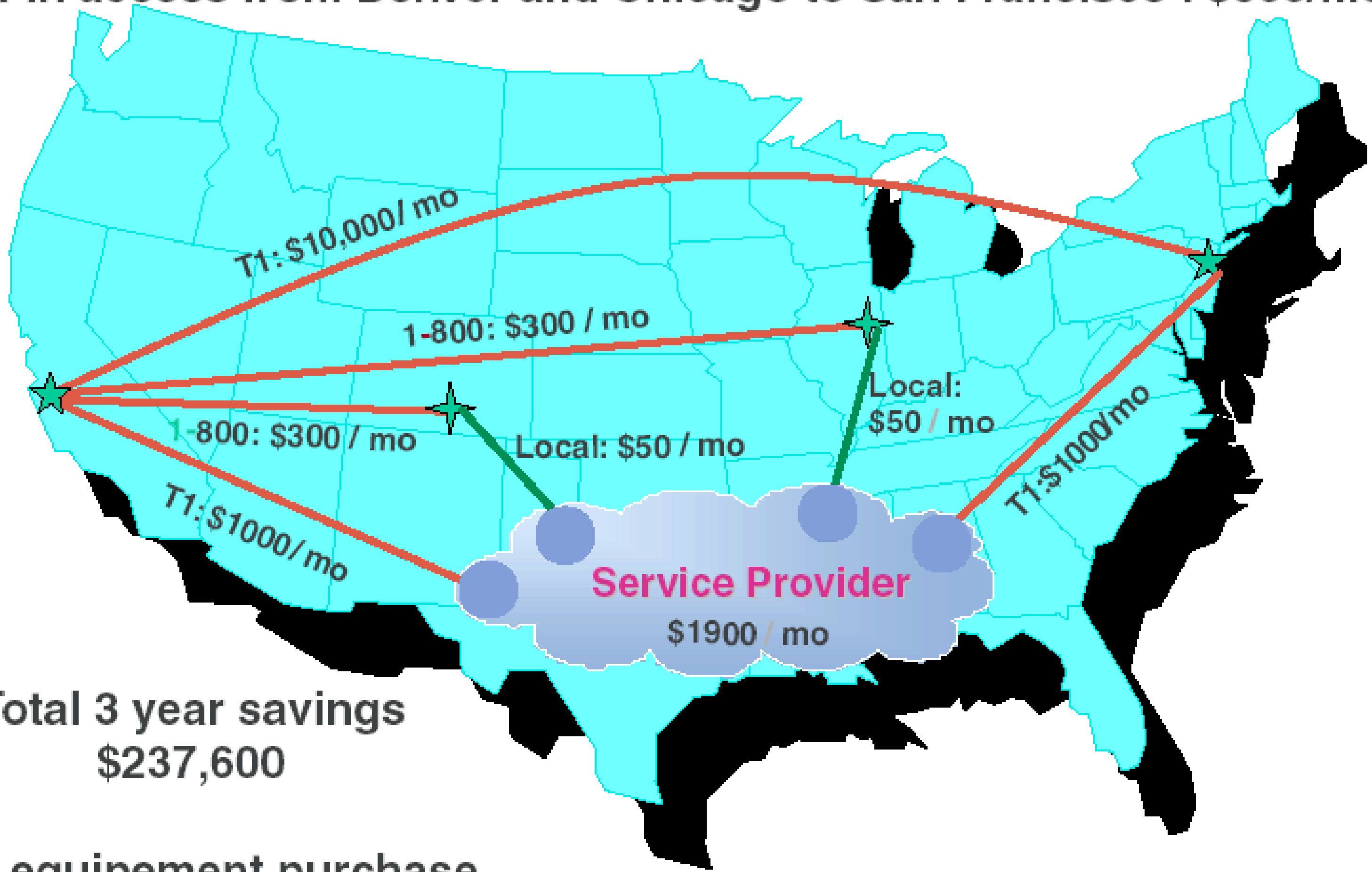
VPNs enable cutting costs with respect to expensive connectivity solutions

Private Networks are based on

- **Private leased lines**
- **Long distance dial-up solutions**

An Example

T1 connections between San Francisco and New York City : $10,000/mo
Dial-in access from Denver and Chicago to San Francisco : $600/mo



Total 3 year savings
$237,600

VPN equipement purchase
$7,800

Why VPN?

VPN enables selective and flexible access to corporate network (services)

- Limited services available to external users
  - High security
  - Few services allowed through firewall
- All intranet functionalities available to corporate users accessing from the Internet
  - VPN connection allowed through firewall
  - Services available as if connected directly to the corporate network

Basic Terminology and Scenarios

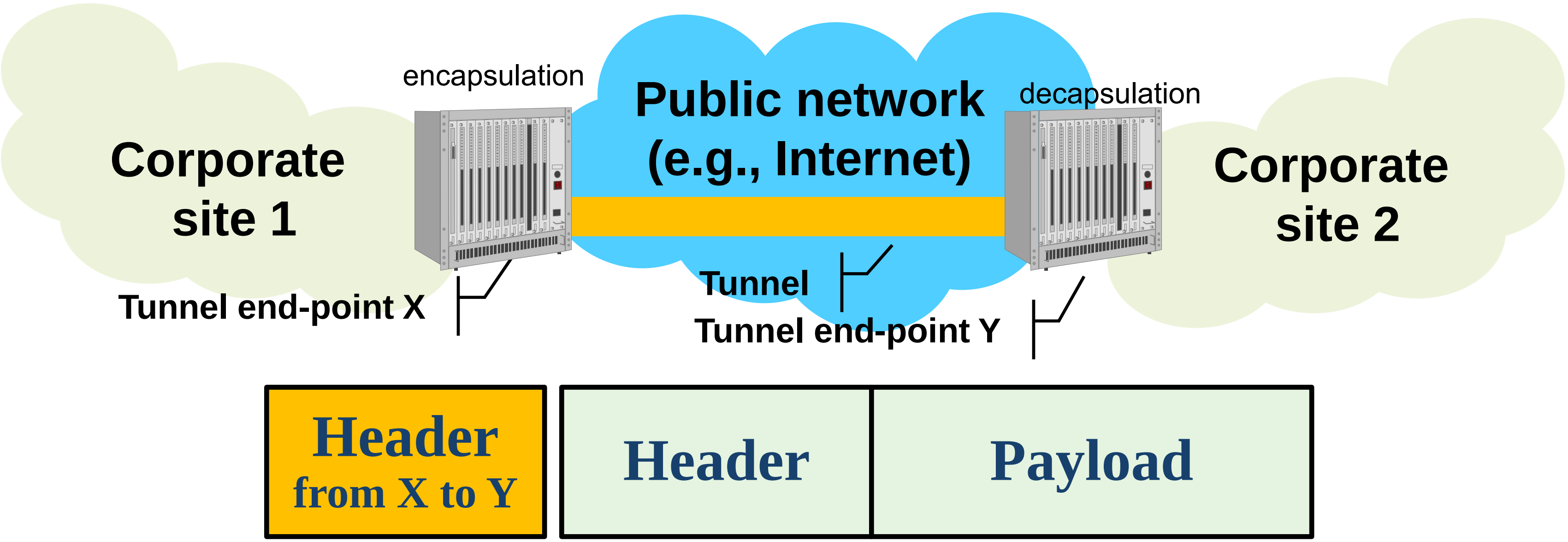
Basic Security Objectives

- End point (e.g., source/destination) authentication
  - Ensure it is what/who it declares to be
- Data integrity no one can modify the data that is sent in the network
  - Ensure data is not changed to prevent tampering
  - (including coming from declared source)
- Data confidentiality no one can read the data
  - Data cannot be accessed/read by anyone else than intended destination
- Data separation we don't want that coca cola sends data to its competitor like pepsi cola.
  - Tunneling

Tunneling

A packet (or frame) between private sites is carried through a public network within a packet handled by public nodes

a packet incapsulated to another packet for a certain part of the path, for example the public network, starting from a corporate site to another corporate site.



S: ip of tunnel end point x
D: ip of end point y.

encripted.
s:ip of a worker in germany
d; ip of polito.

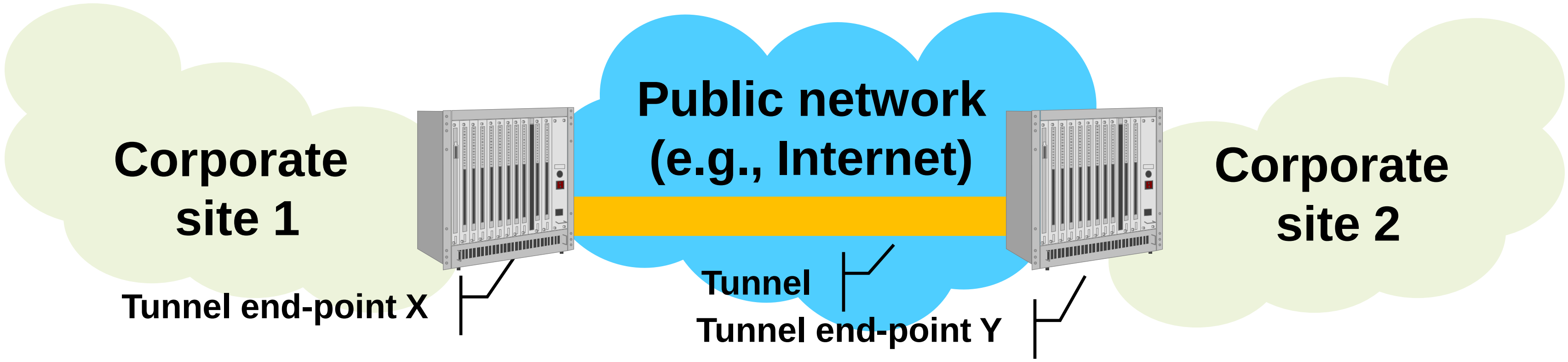
VPN Flavors

- Site-to-site VPN
  - Connect remote networks
  - Virtualizes leased line
- End-to-End VPN
  - Connect remote hosts
  - Virtualizes leased line
- Access VPN or remote VPN or virtual dial in
  - Connects terminal to remote network
  - Virtualizes (dial-up) access connection

s2s VPN Tunneling

- **Tunnel terminated on gateway**

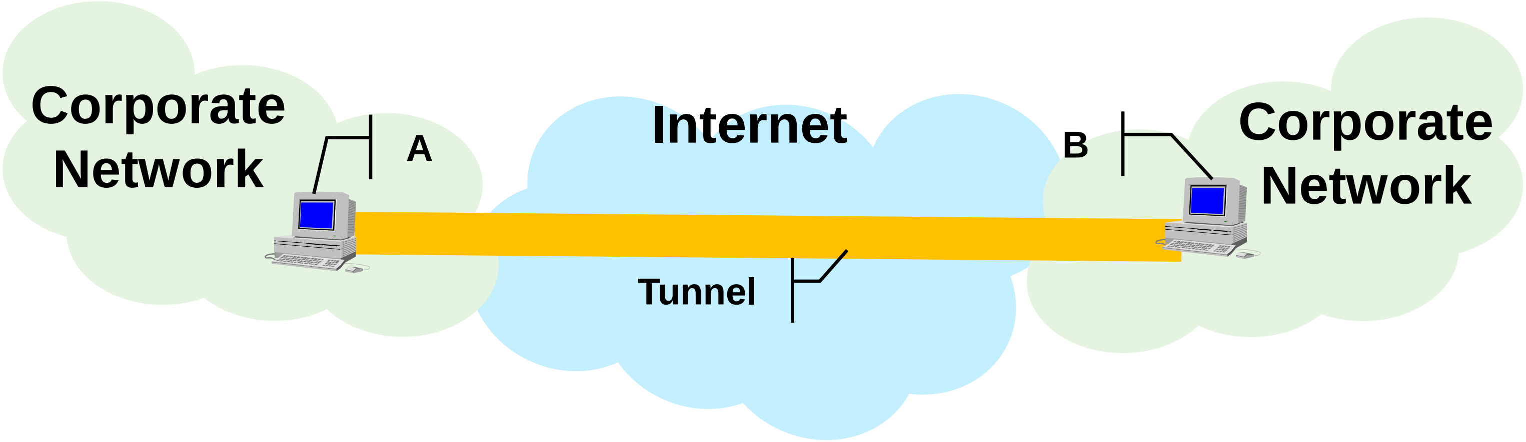
in this example two corporate site exchange data trough public network, thanks to a s2s VPN tunnel.



e2e VPN Tunneling

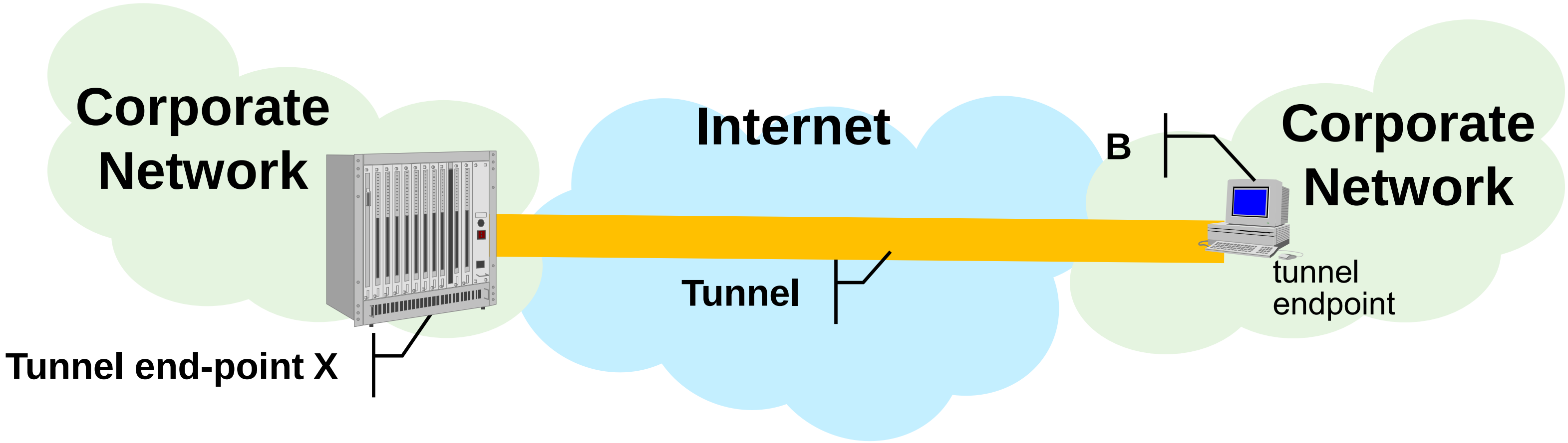
instaed of having vpn gateways , that act as tunnel endpoins,
the tunnel endpoint's are the source and the destination

- Tunnel terminated on end systems



remote VPN Tunneling

- Tunnel terminated on end system and vpn gateway

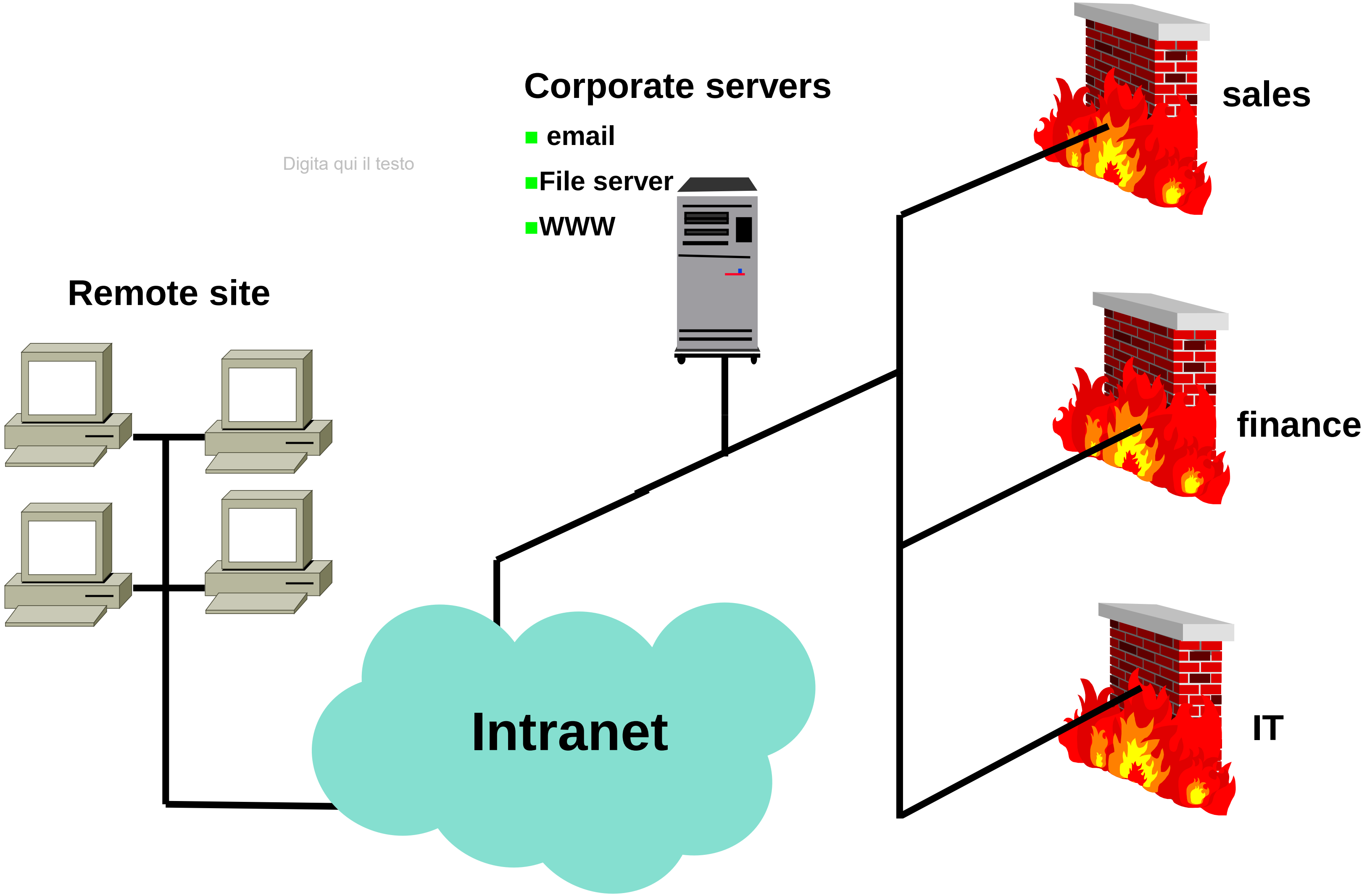


VPN Scenarios

- **Intranet VPN** the intranet is one corporate network that have different sites and we need to interconnect them trough a shared infrastructures that can be the internet.
  - **Interconnection of corporate headquarters, remote offices, branch offices, telecommuter, traveling employee**
- **Extranet VPN**
  - **Interconnection of customers, suppliers, partners, or communities of interest to a corporate intranet**
  - **Provide controlled access to an individual customer/partner/provider user**

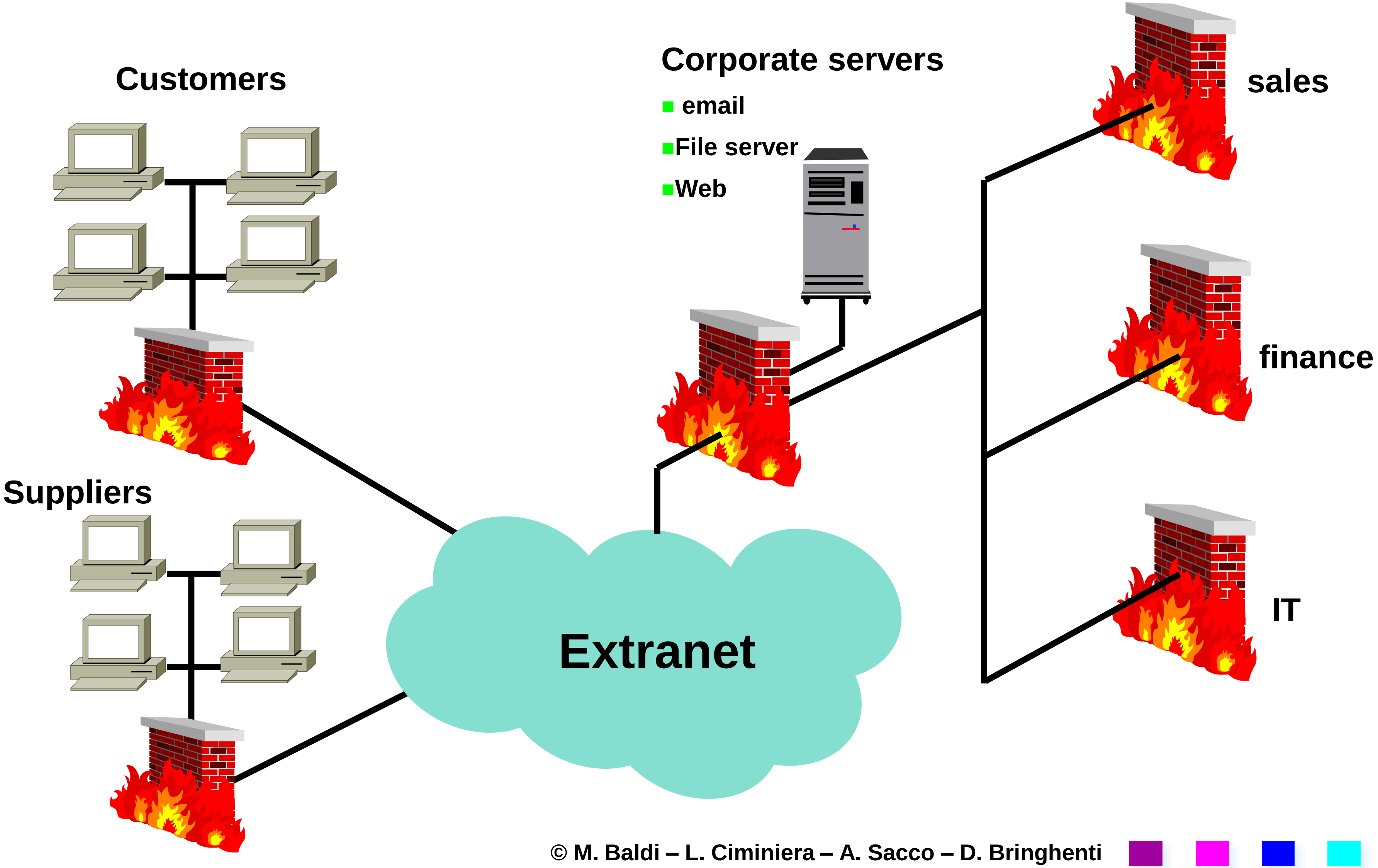
the extranet is a network that is private, but not completely, so for example you have different company that need to interconnect their networks, to join certain services that are not available to the general public.
we use the same protocols, but than we introduce more firewalls that gave us only certain services instaed of all, or of any of them.

Sample Intranet Architecture



when certain level of access had to be garanteed.

Sample Extranet Architecture



Extranet Specific Issues

- *Restricted access to network resources from interconnected networks*
  - Firewall at the VPN
- *Overlapping Address Spaces* private address can be the same for both the companies that are collaborating.
  - Network address translation we can solve this problem with the use of NAT.
- *Open, standard-based solution*
  - Enables interoperability among different organizations
- *Traffic control*
  - Avoid that partner traffic compromises performance on corporate network

how we deal with internet acces?
those host in the vpn, how they will access to internet services.?

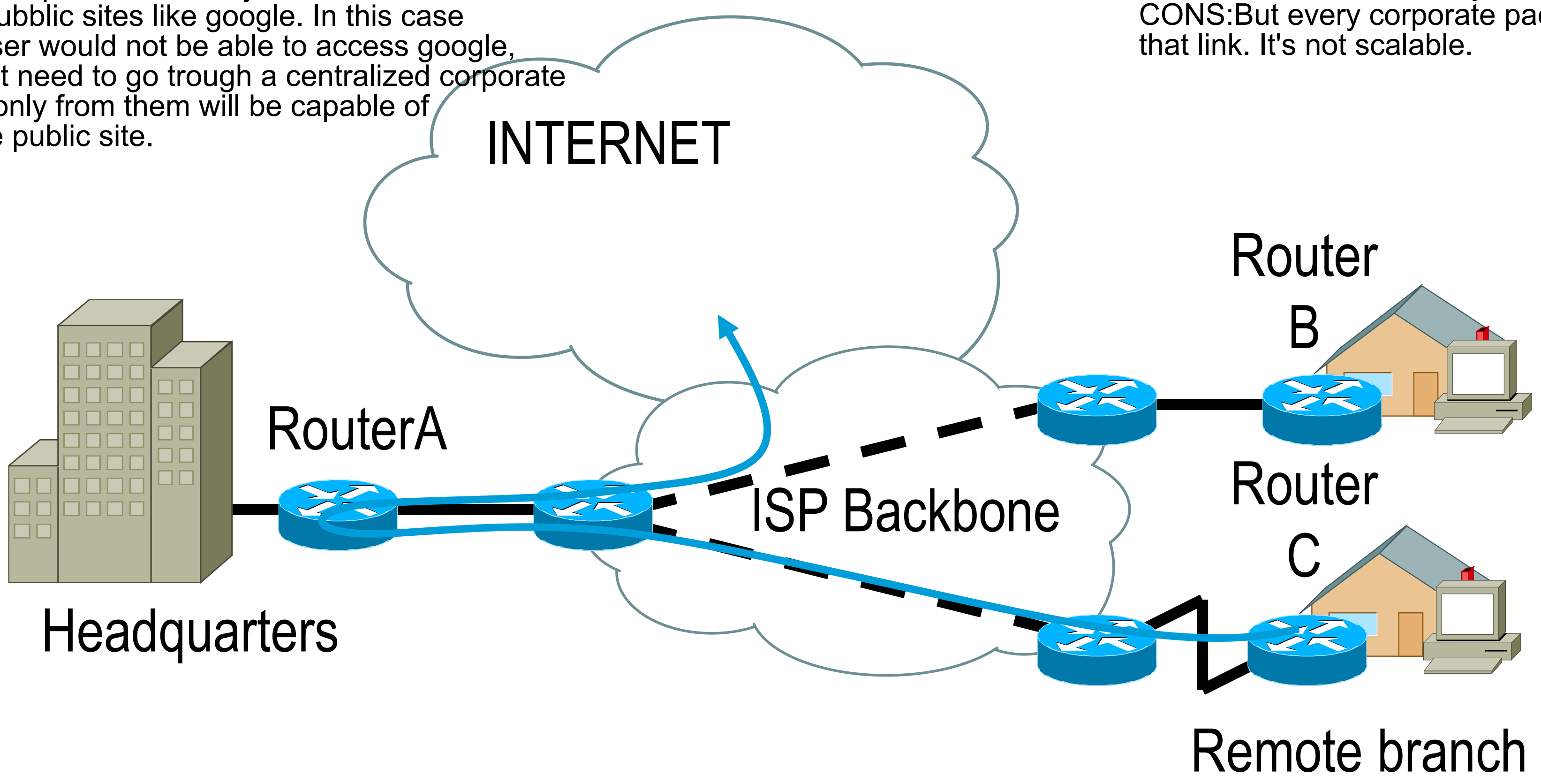
Internet Access

- *Centralized*
  - Remote branches/users use public IP network only to reach headquartes
  - Internet access only from headquarters
  - VPN carries also traffic to and from the Internet
  - Centralized access control
    - Firewall
- *Distributed (voluntary connection)*
  - Remote branchs/users access the Internet through their IP network connection
  - VPN is deployed only for corporate traffic

Centralized Internet Access

if i have a remote user, working somewhere at home, or a remote group, and then they want to access, some other pubblic sites like google. In this case the remote user would not be able to access google, but the packet need to go trough a centralized corporate network and only from them will be capable of accessing the public site.

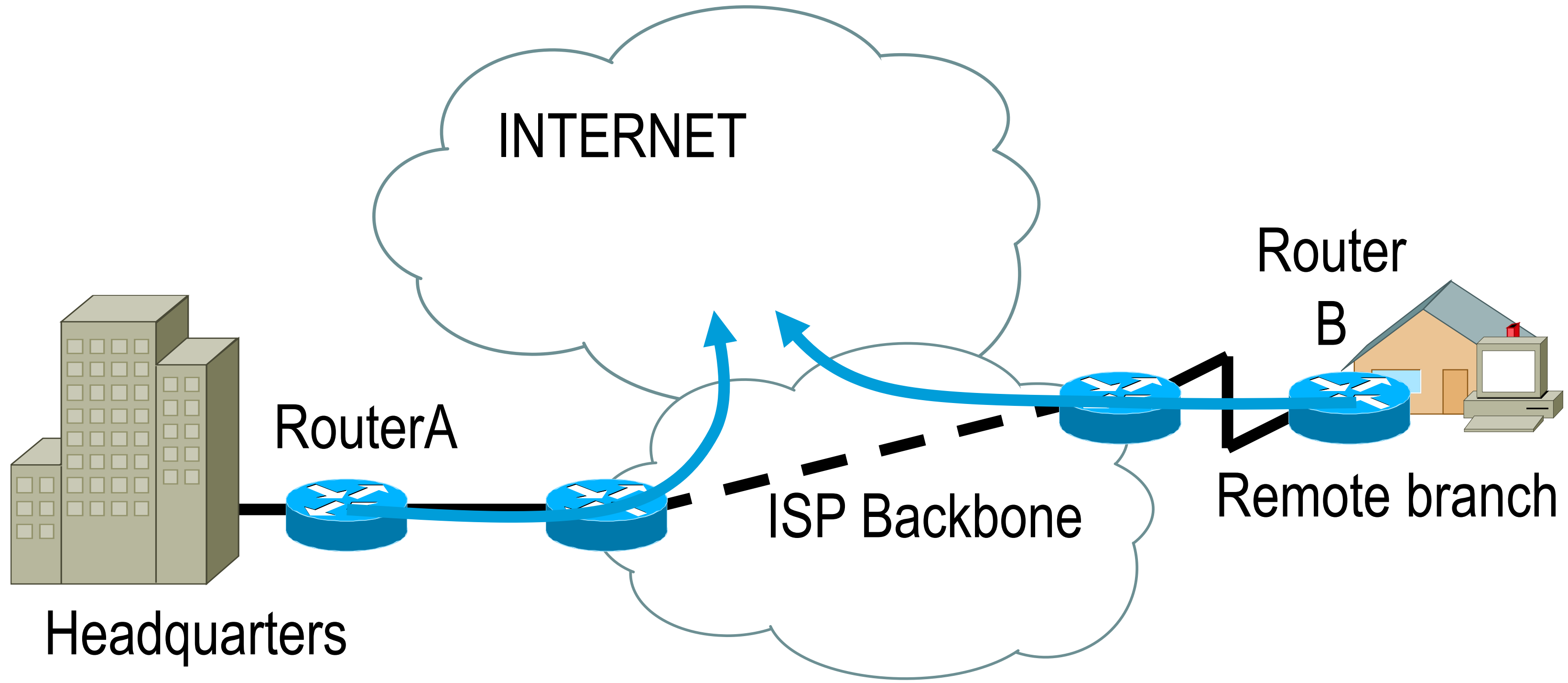
PRO:only one firewall before the headquarters.
We cannot create security breaches.
CONS:But every corporate packet has to go trough that link. It's not scalable.



Distributed Internet Access

we allow directly the remote user to access the internet immediately with a remote branch.

one firewall (because we want to avoid unwanted traffic)for each corporate site,
there could be security breaches,
but is a lot more scalable, by definition.



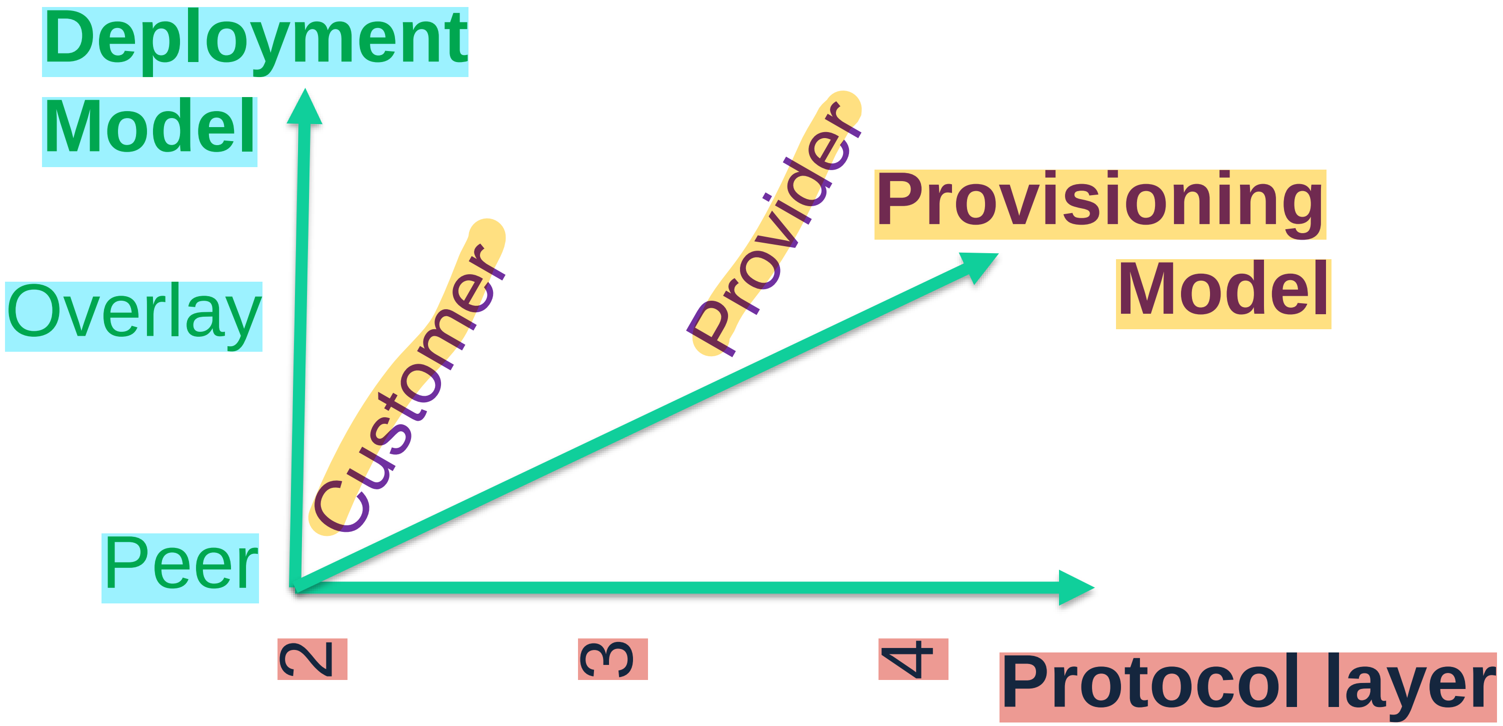
(Virtual) VPN Topologies

- Hub and spoke
  - Each branch communicates directly with headquarters
  - Fits to data flow of many corporations
    - Mainframe or data-center centered
  - Routing is sub-optimal
  - Small number of tunnels
  - Hub could become bottleneck
- Mesh
  - Larger number of tunnels
    - Harder to manually configure
  - Optimized routing

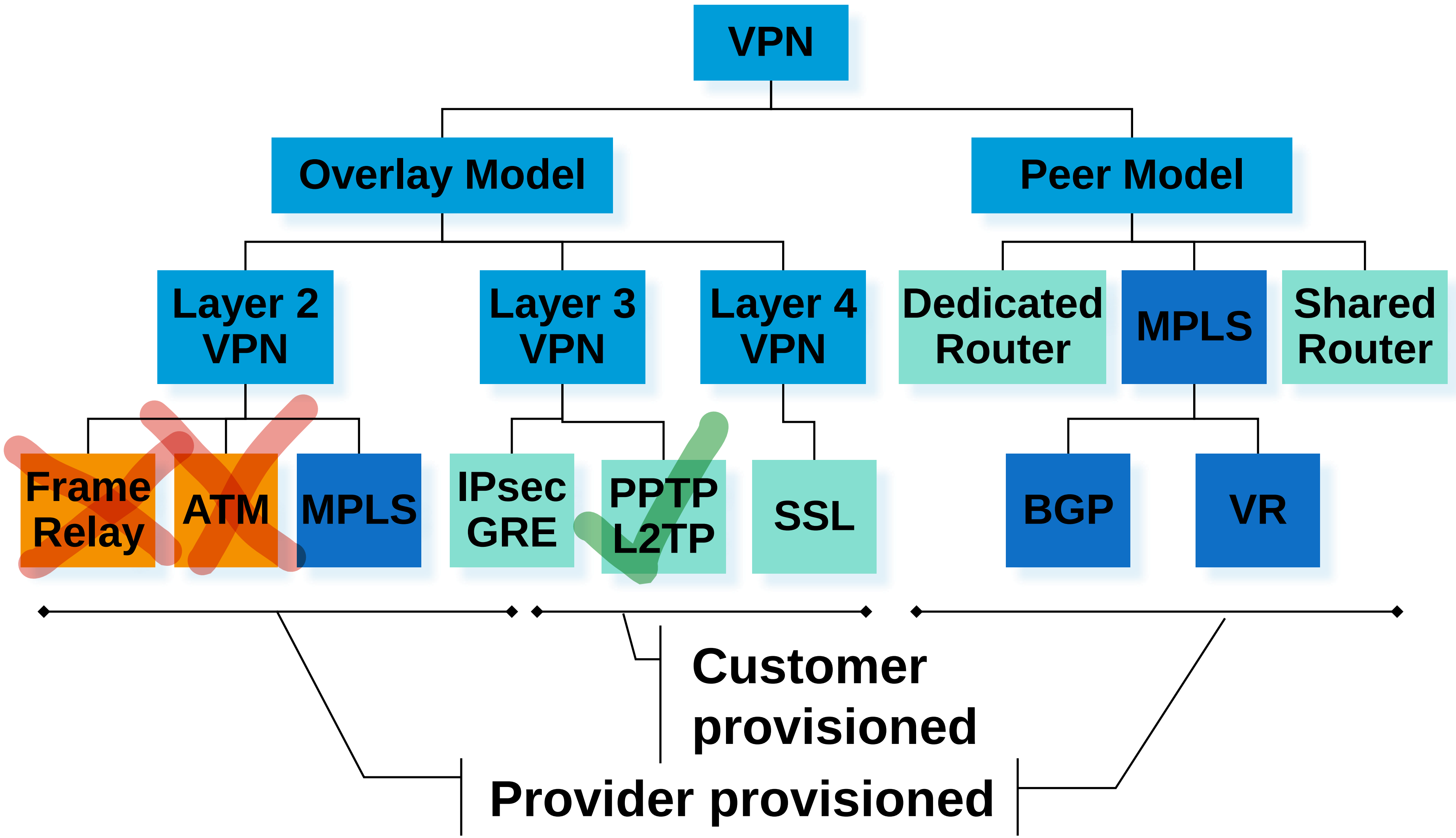
Deployment Models, Provisioning Models & VPN Protocol Layers

Many VPN solutions:
let's try to identify key features

Three dimensions



And Categorize the Many VPN solutions



Overlay Model

little networks. Less expansive

- The public network does not participate in realizing the VPN
  - It does not know where VPN destinations are
  - Just connectivity among VPN gateways
- Each VPN gateway must be “in touch” with every other VPN gateway
  - E.g., highly meshed tunnels
- Routing is performed by the VPN gateways

Peer Model

more scalable solutions

internet knows about the VPN

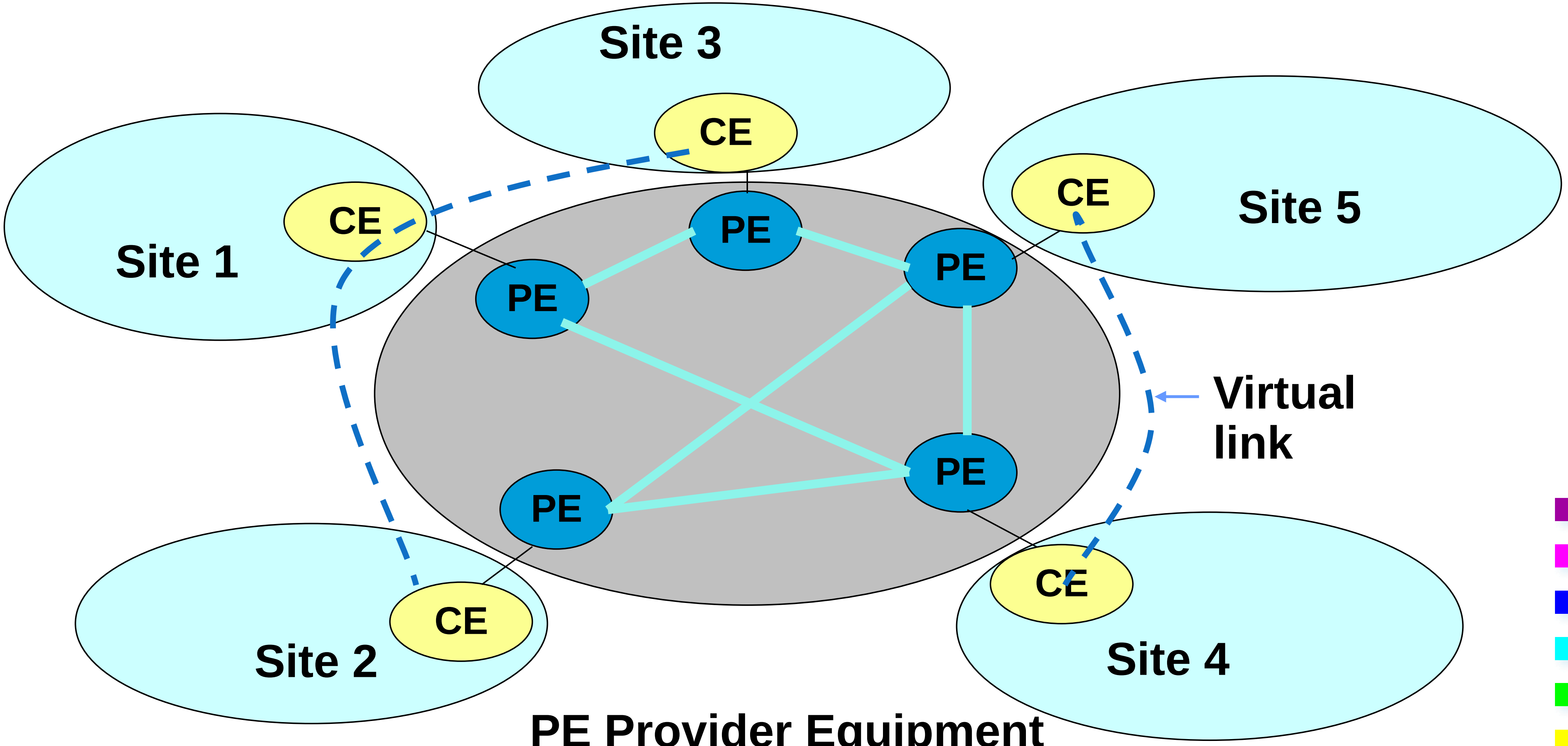
- Each VPN gateway interacts with a public router (its peer)
  - Exchange of routing information
  - Service provider network disseminates routing information
- Public network routes traffic between gateways of the same VPN

Customer Provisioned VPN

- Customer implements VPN solution
  - Owns, configures, manages devices implementing VPN functionalities
    - Customer equipment
- Network provider is not aware that the traffic generated by customer is VPN
- All VPN features implemented in customer devices
- CE terminates tunnels

Customer Provisioned VPN

usually terminate tunnels.



PE Provider Equipment

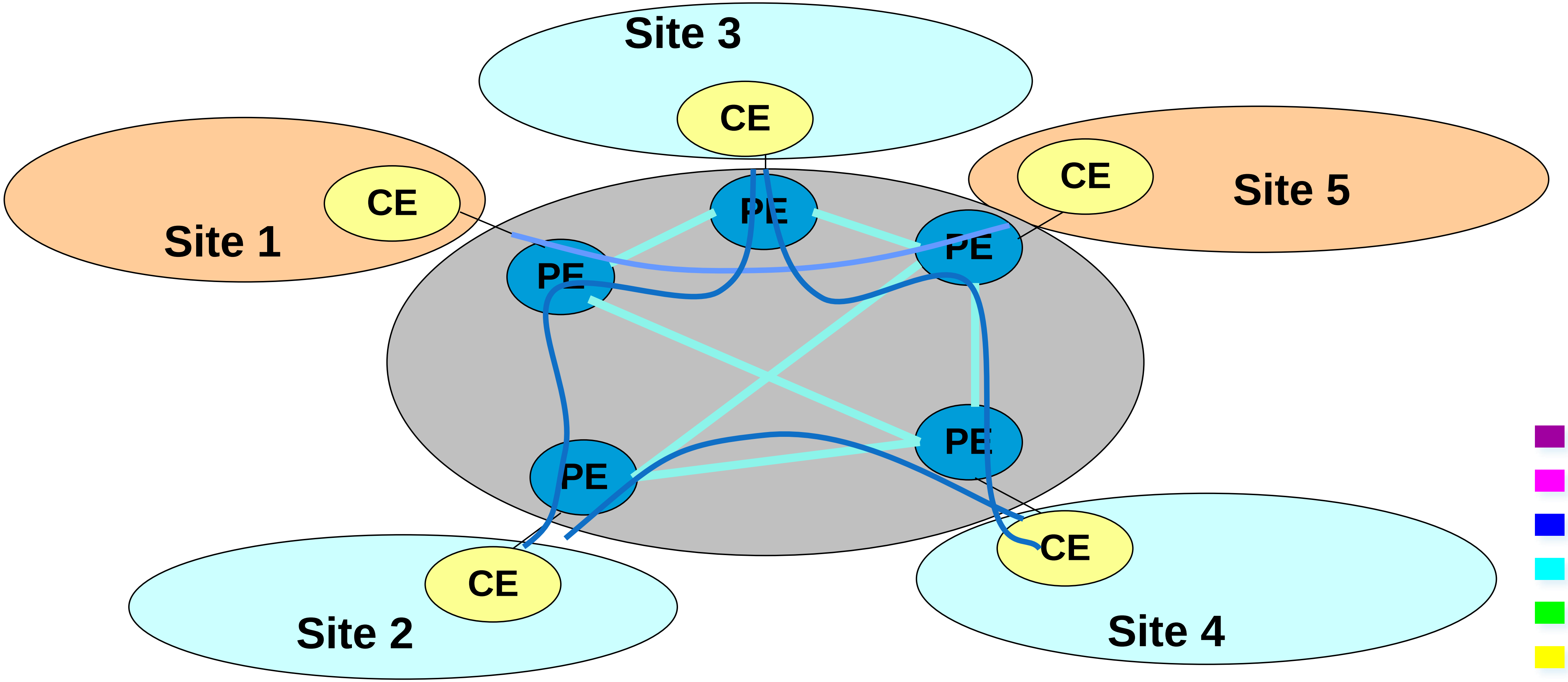
CE Customer Equipment

Provider Provisioned VPN

- Provider implements VPN solution
  - Owns, configures, manages devices implementing VPN functionalities
- VPN state maintained by the provider devices
- Traffic belonging to different VPNs is separated by the provider devices
- CE may behave as if it were connected to a private network
- PE terminates tunnels

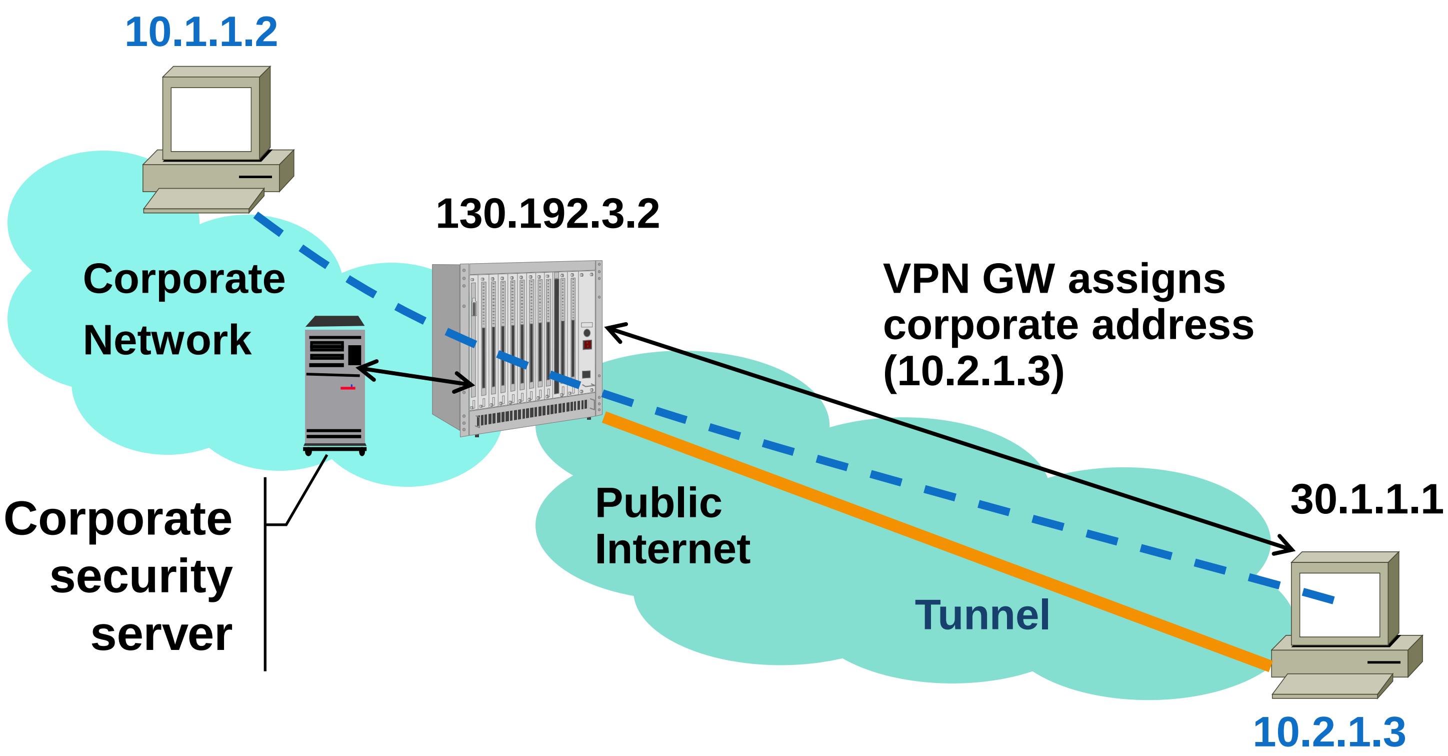
Provider Provisioned VPN

customer has to know nothing, except for trusting the service provider.



Access VPN Customer Provisioned

vpn where remote host is a VPN gateway endpoint.



Tunnel endpoints: 130.192.3.2 – 30.1.1.1

Connection endpoints: 10.2.1.3 – 10.1.1.2

Customer Provisioned vs. Provider Provisioned

- Customer Provisioned
  - Remote host has 2 addresses
    - ISP assigned and corporate
  - Remote host terminates VPN tunnel
  - Remote host must activate tunnel
    - If tunnel is not activated, client can operate without VPN
  - Can be used from any Internet connection (ISP)
- Provider provisioned
  - Remote host has 1 address (corporate)
  - NAS terminates VPN tunnel
  - Remote host is always on VPN
  - Internet access is only centralized
  - Requires access to specific ISP

Layer N VPN

Packet transport (tunneling) provided

- **by Layer N Protocol**

and/or

- **as Layer N service**

layer 3 vpn, the package is encapsulated in the layer 3, the IP level., or because is using a layer 3 protocol (slide 39).

Layer 2 VPNs

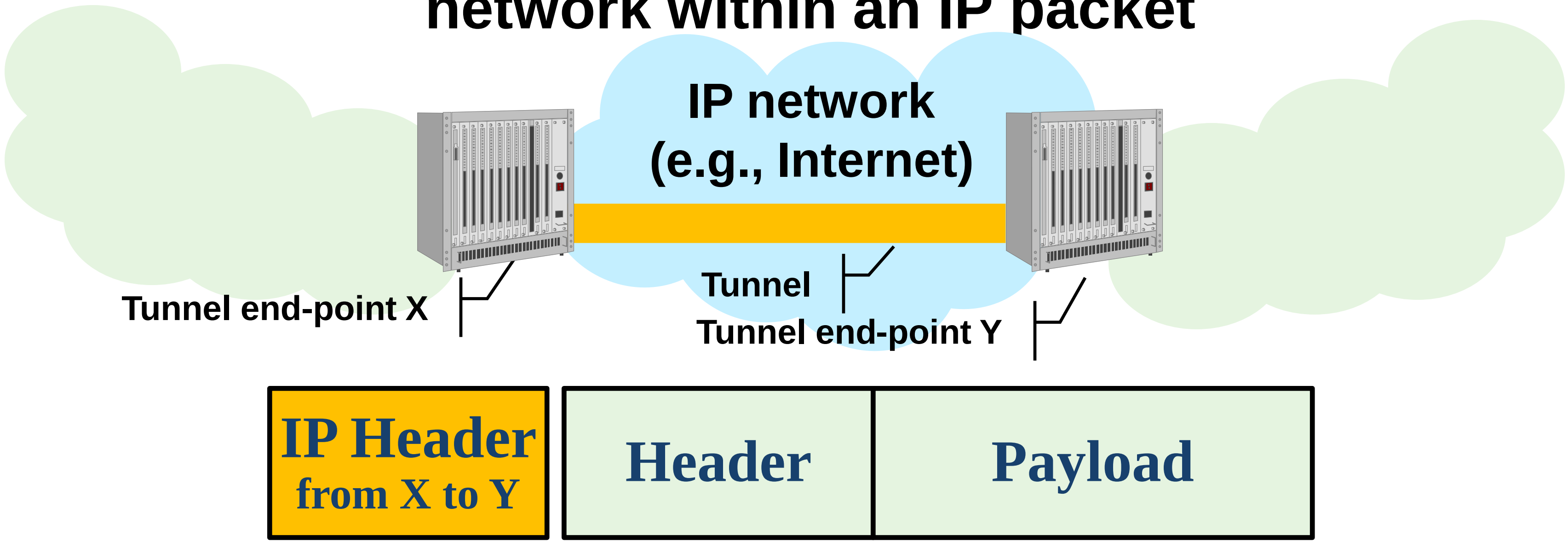
- *Virtual Private LAN Service*
  - Emulates functionalities of LANs
  - Can be used to connect LAN segments
    - Works as single LAN through the public network
  - VPN solution emulates learning bridges
    - Routing based on MAC addresses
- *Virtual Private Wire Service*
  - Emulates a leased line
  - Any protocol can be carried
- *IP-Only LAN-like Service*
  - CEs are IP routers or IP hosts (not Ethernet switches)
  - Only IP (plus ICMP and ARP) packets travel through the VPN

Layer 3 VPNs

- Layer 3 packets are forwarded through the public network
- Routing based on layer 3 addresses
  - Peer: VPN/corporate/customer addresses
  - Overlay: backbone addresses
- CEs are either IP routers or IP hosts

Tunneling in Layer 3 VPN

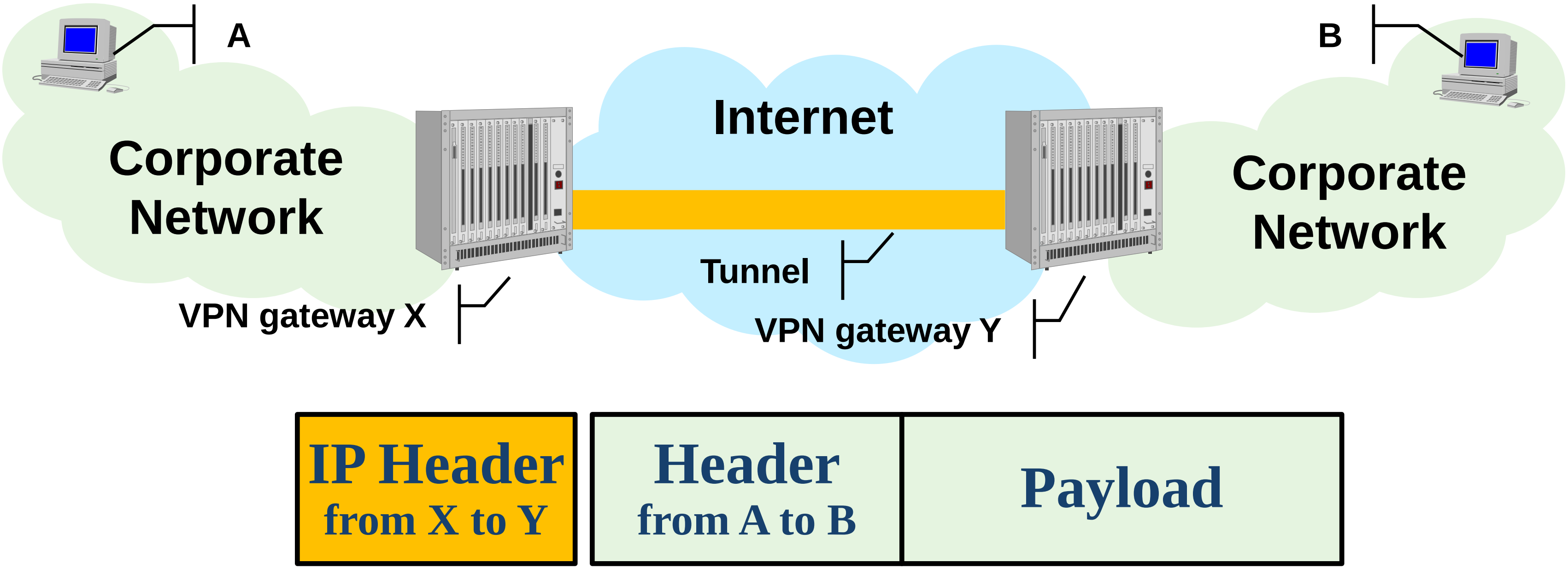
A packet (or frame) is carried through an IP network within an IP packet



- An IP packet within an IP packet (IP-in-IP)
  - GRE, IPsec
- A layer 2 frame, within an IP packet
  - L2TP, PPTP (based on GRE)

IP in IP Tunneling:

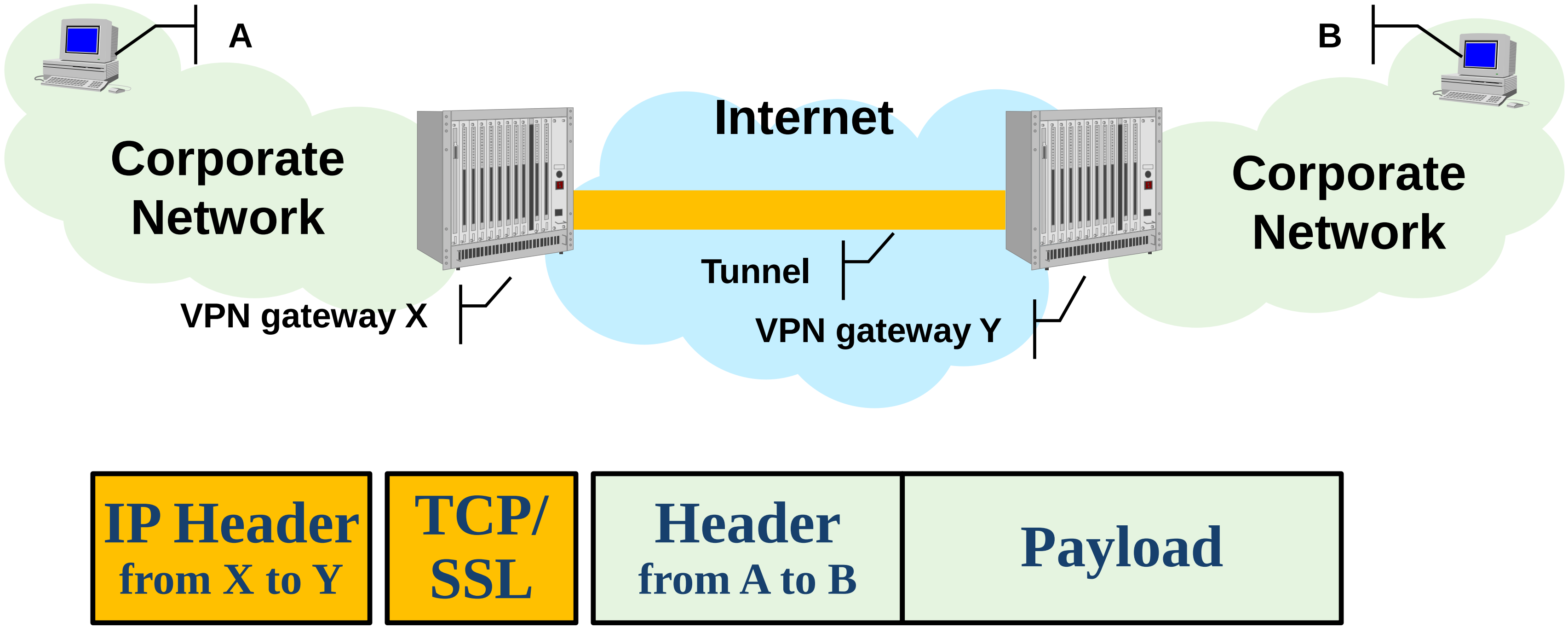
- A and B are corporate addresses
  - Not necessarily public
- Tunneling enables communication
- Tunneling by itself does not ensure security



Layout 4 VPN Tunneling: s2s

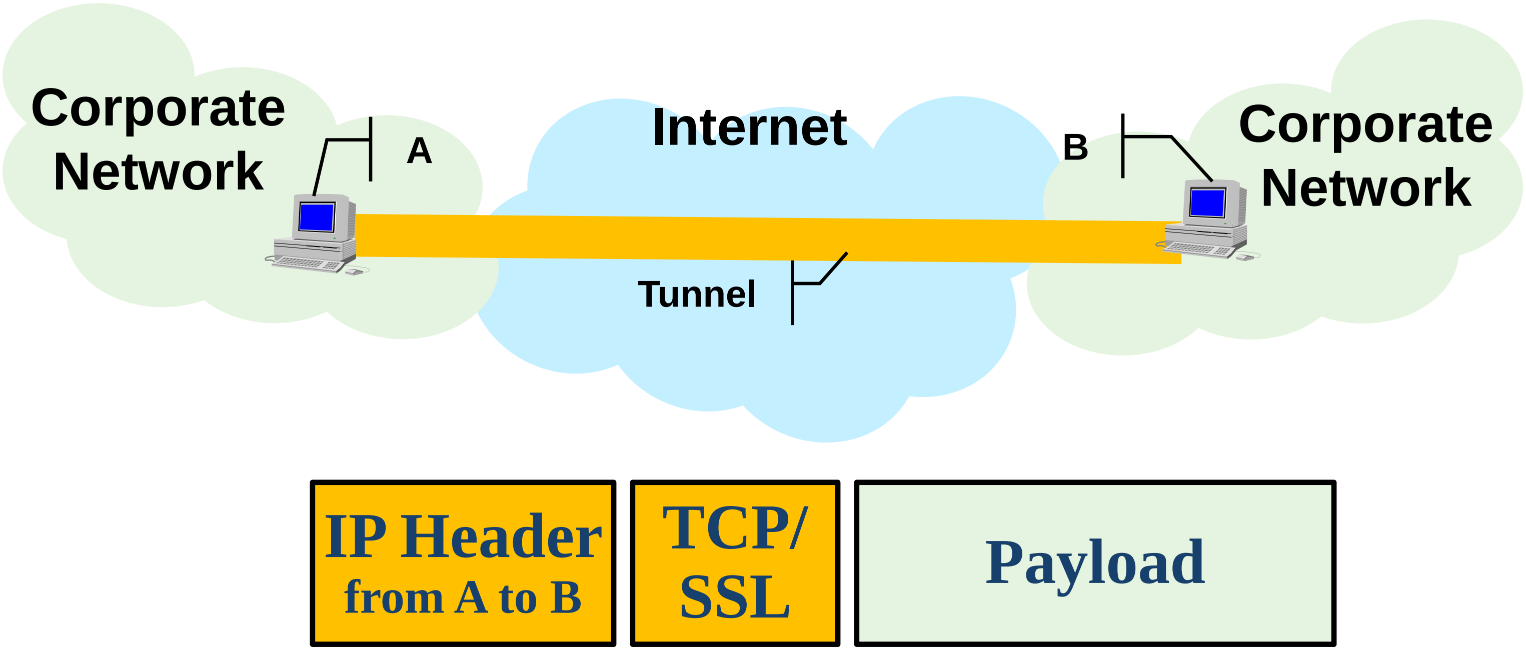
- VPN built using TCP connections
  - Tunnels realized by TCP connections
- Security achieved with SSL/TLS

frame of layer 2 encapsulated into a layer 4 packet
any other occurrencies can happen.

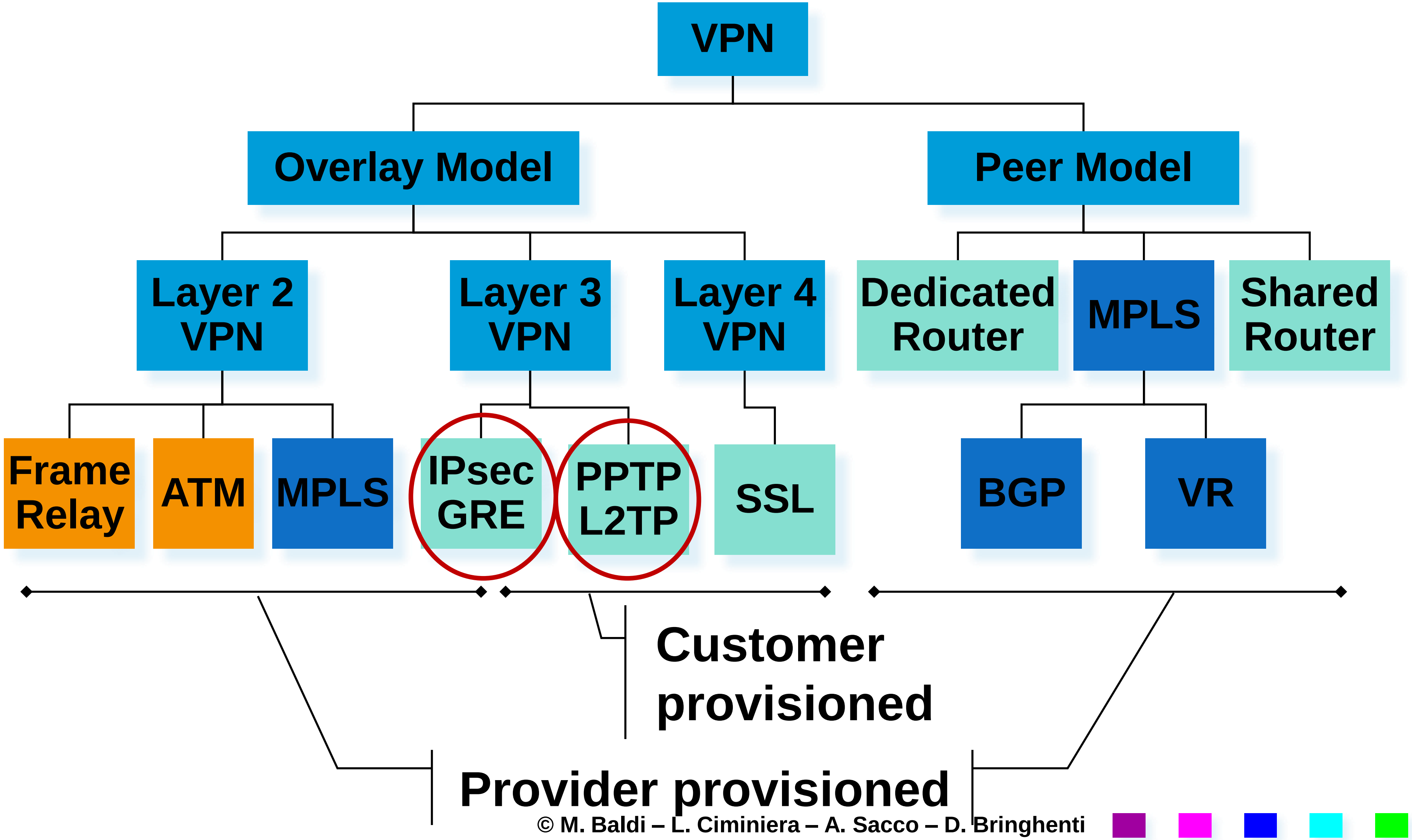


Layer 4 VPN Tunneling: e2e

- Tunnel possibly terminated on end systems



And Categorize the Many VPN solutions



GRE = tunneling into IP packet.

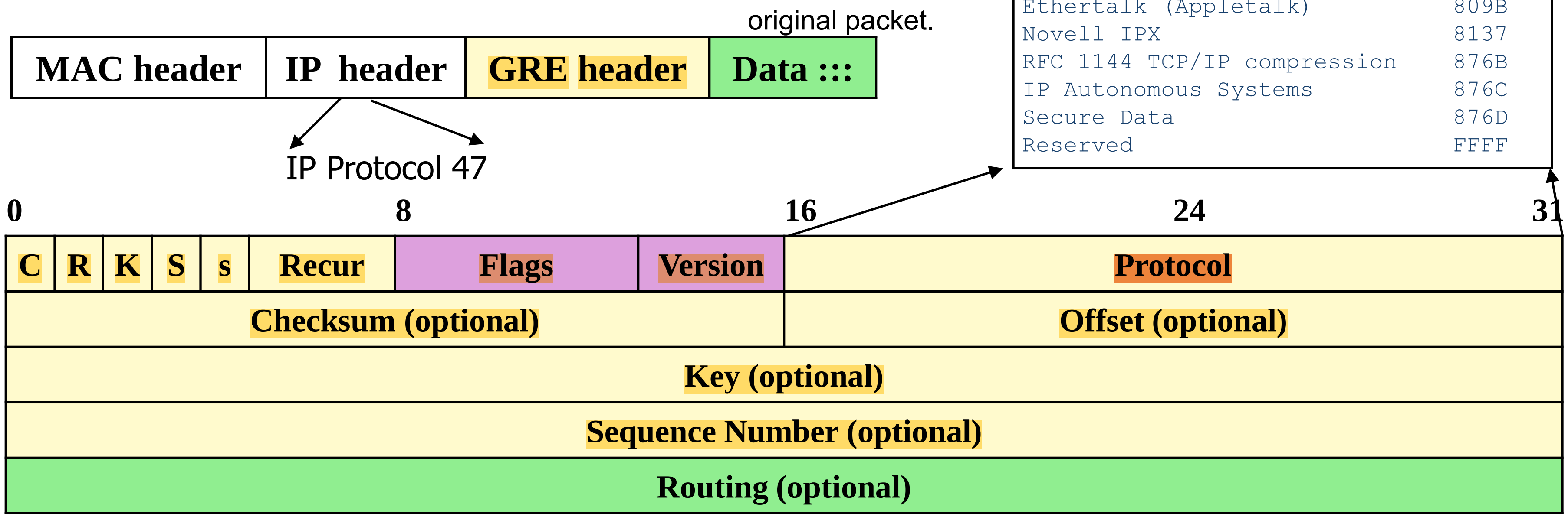
GRE – Generic Routing Encapsulation

not designed for VPN, but designed for tunneling.

Packet Format

- **Encapsulation (tunneling) of any protocol (including IP) into IP**

- **Header version 0**



Protocol Family	PTYPE
Reserved	0000
SNA	0004
OSI network layer	00FE
PUP	0200
XNS	0600
IP	0800
Chaos	0804
RFC 826 ARP	0806
Frame Relay ARP	0808
VINES	0BAD
VINES Echo	0BAE
VINES Loopback	0BAF
DECnet (Phase IV)	6003
Transparent Ethernet Bridging	6558
Raw Frame Relay	6559
Apollo Domain	8019
Ethertalk (Appletalk)	809B
Novell IPX	8137
RFC 1144 TCP/IP compression	876B
IP Autonomous Systems	876C
Secure Data	876D
Reserved	FFFF

Header fields

- C, R, K, S
  - Flags indicating the presence/absence of optional fields
- s
  - Strict source routing flag
  - if the destination is not reached when the source route list end, the packet is dropped
- Recur
  - Max. number of additional encapsulation permitted (must be 0)
- Protocol
  - ID of the payload protocol
- Routing
  - Sequence of router IP addresses or ASs for source routing

Enhanced GRE (version 1)

- Deployed by PPTP
- `Acknowledgment Number`
  - Delivery of packets by remote end-point can be notified

ack. flag

<table>
<tr><td>0</td><td></td><td></td><td></td><td></td><td></td><td>8</td><td></td><td></td><td>16   24   31</td></tr>
<tr><td>C</td><td>R</td><td>K</td><td>S</td><td>s</td><td>Recur</td><td>A</td><td>Flags</td><td>1 (ver)</td><td>Protocol Type</td></tr>
<tr><td colspan="9">Key (HW) Payload Length</td><td>Key (LW) Call ID</td></tr>
<tr><td colspan="10">Sequence Number (Optional)</td></tr>
<tr><td colspan="10">Acknowledgment Number (Optional)</td></tr>
</table>

Advanced Functionalities

- *Key (high 16 bit)*
  - Payload length: no. of bytes excluding GRE header
- *Key (low 16 bit)*
  - Call ID: session ID for this packet
- *Sequence number*
  - For ordered delivery, error detection and correction
- *Acknowledge number*
  - Highest number of GRE packet received in sequence for this session
    - Cumulative ack

Other mechanisms in GRE

- *Flow control*
  - Sliding window mechanism
- *Out-of-order packets*
  - Discarded, because PPP allows lost packets, but cannot handle out-of-order packets
- *Timeout values*
  - Re-computed each time an ack packet is received
- *Congestion control*
  - Timeouts do not cause re-transmission
    - Used only to move sliding window
    - Packets will be lost
  - Their value should be rapidly increased

Layer 2 frame, within an IP packet

Two Protocols for Access VPN

- L2TP (Layer 2 Tunneling Protocol)
  - Initially only Provider provisioner
    - (Initially) not widely implemented in terminals
  - Idependent of layer 2 protocol on host
  - Security through IPsec
    - Strong, but complicated

implemented in the software of the modern hosts, directly integrated on OS.

- PPTP (Point-to-Point Tunneling Protocol)
  - Castomer provisioner
  - Originally proposed by Microsoft, Apple, …
  - Weak encryption and authentication
  - Proprietary key management

L2TP- Layer 2 Tunneling Protocol

Solution Components

- Tunneling between public network access point and corporate network
  - Also wholesale dial-up services
    - Between access provider and Internet Service Provider
- L2TP Access Concentrator (LAC)
  - Network access device supporting L2TP
  - NAS (Network Access Server)
- L2TP Network Server (LNS)
  - Corporate (VPN) Gateway
- Customer provisioned deployment mode by including LAC functionality in host

The goal of access vpn is to allow a remote host, to act as if it were on the corporate network

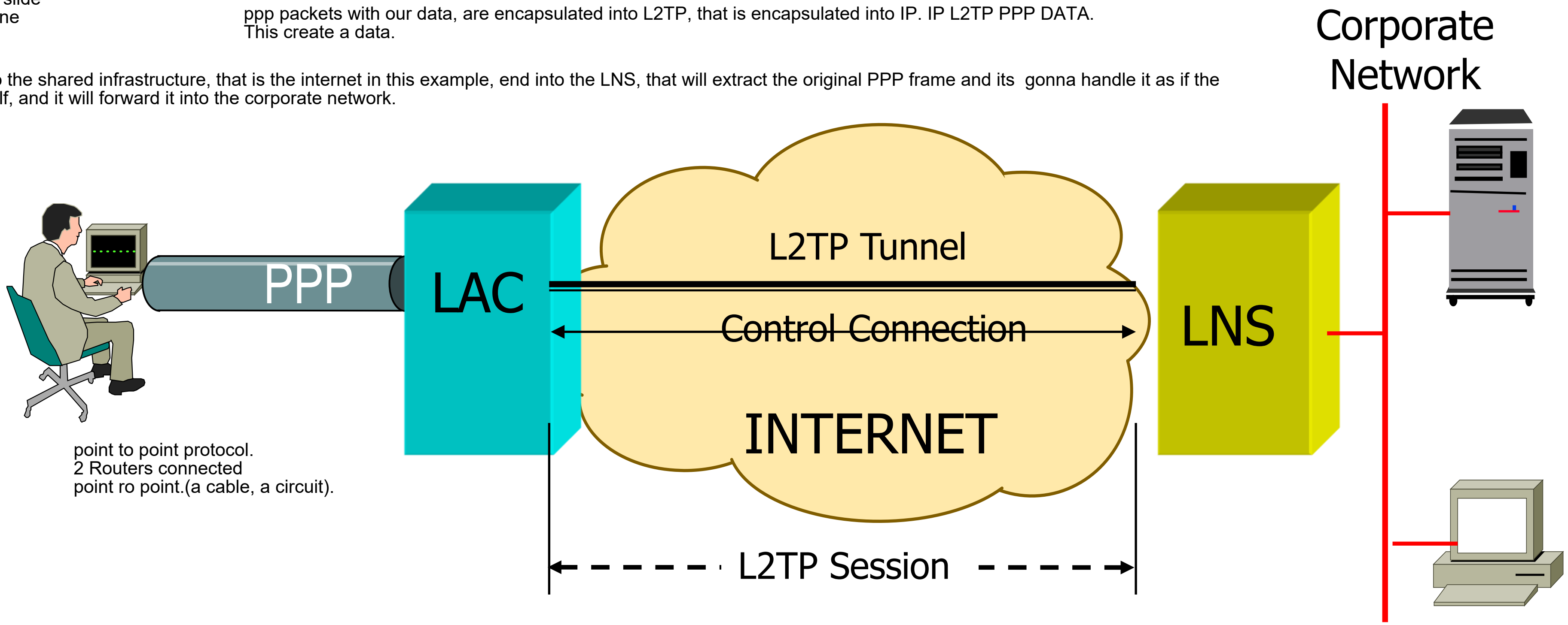
the goal of the access vpn is to allow a remote host to act as if it were on the corporate network,
which means having the same address it would have on the corporate network, and consequently
getting access to services that are allowed only on that corporate address.
In the meantime we want to ensure some security features, like data integrity, data confidentiality and identification of the Endpoints.
How we do this with L2TP(Layer 2 Tunneling Protocol).
is a protocol that allows us to take layer 2 frames,in this case ppp frames that goes from the host to an access device in which the host is connected (LAC, Lt2p access concentrator).

Every frame that arrives into LAC, will be encapsulated, from the LAC, to the LNS.
The protocol that we use to do this tunnel is l2tp protocol. Vedi slide lt2p header per continuazione

Original Reference Scenario

ppp packets with our data, are encapsulated into L2TP, that is encapsulated into IP. IP L2TP PPP DATA.
This create a data.

once the packet travel into the shared infrastructure, that is the internet in this example, end into the LNS, that will extract the original PPP frame and its gonna handle it as if the host was attached to itsself, and it will forward it into the corporate network.



point to point protocol.
2 Routers connected
point ro point.(a cable, a circuit).

Provider provisioned deployment mode

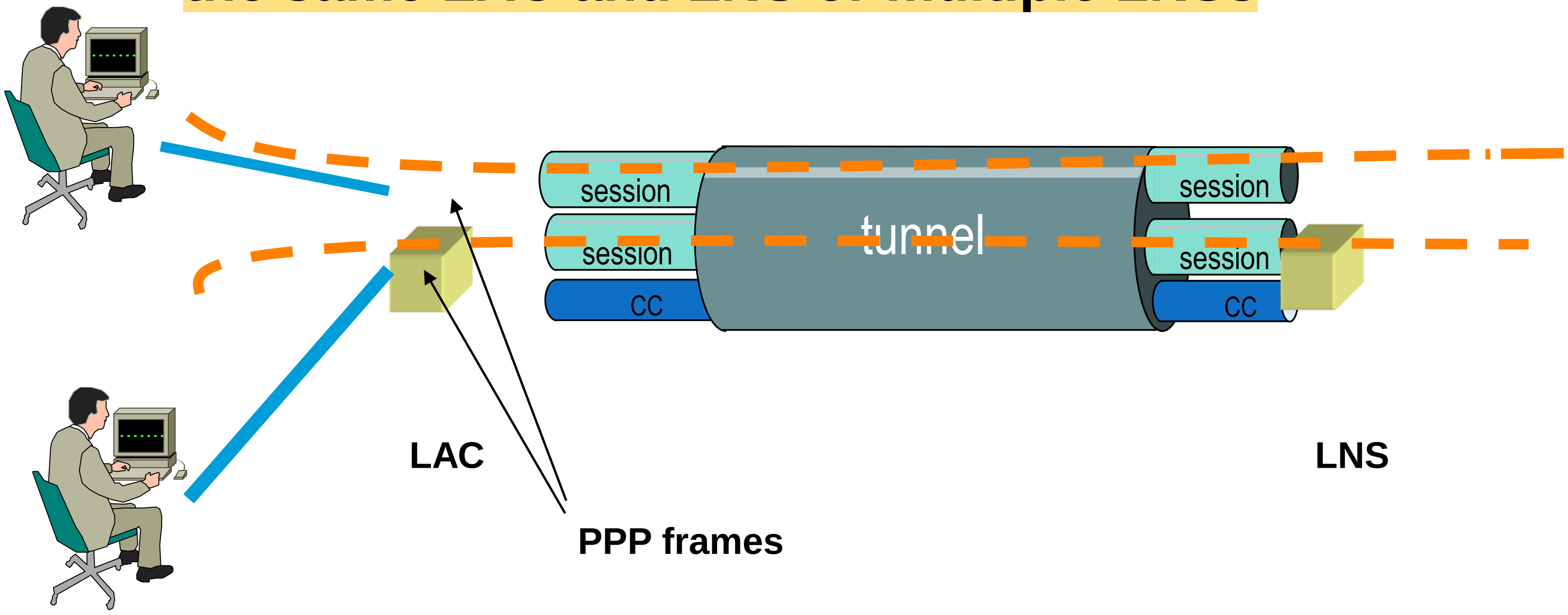
provider offer a VPN solution. The provider is managing LAC and LNS(L2TP Network Service)

CHAP
1. **Access**: The user wants to connect to the VPN through the **LAC**.
2. **Challenge**: The server (**LNS**) sends a random number to the LAC as a "challenge."
3. **Response**: The LAC combines the challenge with the password and sends a secret code (a hash) to the LNS, without transmitting the actual password.
4. **Verification**: The LNS calculates the same code; if it matches, the connection is authorized.
5. **Security**: The password is never directly sent, making the connection more secure.

anytime that the host wants to connect the LAC will send a new challenge(that is another random sequence of bytes) to the user.

Tunnels and sessions

- Multiple sessions may exist within the same tunnel
- Multiple tunnels may be established between the same LAC and LNS or multiple LNSs



L2TP Operation

1. Establish a control connection for a tunnel between LAC and LNS
2. Establish one or more sessions triggered by a call request

- The control connection must be established before a connection request is generated
- A session must be established before tunnelling PPP frames

Establishing a tunnel

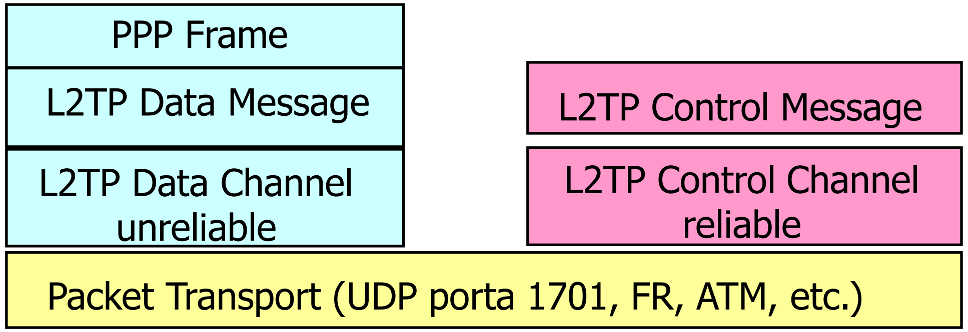
- Peer can be authenticated
- A shared secret must exist between LAC and LNS
- L2TP uses a CHAP-like mechanism
  - Challenge-Handshake Authentication Protocol
  - A challenge is proposed to the other peer
  - The correct answer to the challenge requires the shared secret
    - Cryptographic algorithm used to create the answer
- The tunnel endpoints exchange the local ID attributed to the tunnel

L2TP Header

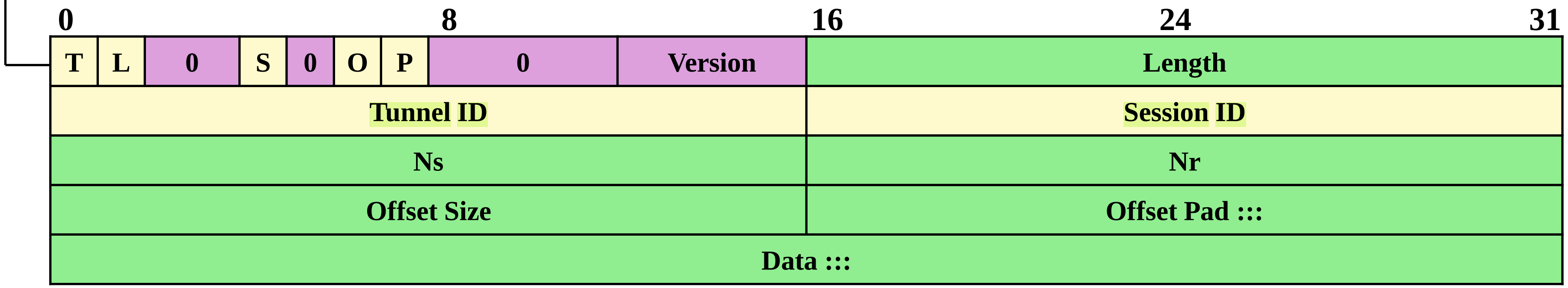
the original packet. a ppp frame, has been encapsulated into a LT2P header, encapsulated into a UDP message, with port 1701, encapsulated into an ip header(with s: LAC and d: LNS).

ritorna a Original refence scenario

- Control Message
- Data Message



T	Description
0	Data message.
1	Control message.



Header fields

- *L, S, O*
  - Flags indicating whether the fields length, Ns & Nr and offset are present
  - For control messages L=S=1 and O=0
- *P*
  - Priority flag, if set, the priority is high
- *Ver*
  - Version, must be 2
- *Tunnel ID*
  - Recipient's ID of the control connection (local meaning)
- *Session ID*
  - Recipient's ID of the session within the same tunnel (local meaning)

Other header fields

- ***Ns***
  - Sequence number of the data or control message
- ***Nr***
  - Sequence number of the next control message to be received (i.e. last Ns received in order +1 modulus 2^{16}
- ***Offset***
  - Number of bytes, past the header, where the payload data starts

Sequence Numbers

- Data connections use sequence numbers only to detect out of order packets
- No re-transmission for data streams
- No ack in data streams
  - Layer 2 protocol, e.g., HDLC, can possibly take care of this
- Control packets use ack and re-transmission
  - Selective repeat
  - Tx and Rx windows set to 32k

Security issues

- *Tunnel endpoint authentication*
  - Authentication only during tunnel establishment phase
  - A user who can snoop traffic, can easily inject packets in a session
  - Tunnel and session IDs should be selected in an unpredictable way (not sequentially)
- *Packet level security*
  - Encryption, authentication, and integrity must be provided by the transport mechanism
    - E.g., IPsec
- *End-to-end authentication*
  - Provided by the transport mechanism (e.g. IPsec)

PPTP – Point-to-Point Tunneling Protocol

The goal of access vpn is to allow a remote host, to act as if it were on the corporate network,

work in a similar way of the L2TP, but this was created by big companies(Apple, Microsoft), so that they could have a solution right into their laptop, without a LAC, and without the intervetion of a provider.

The LAC functionality is embedded inside the end hosts, so that we have not the exchange of the information between the user and the lac, that we had in the L2TP. everything happens inside the host in the network stack

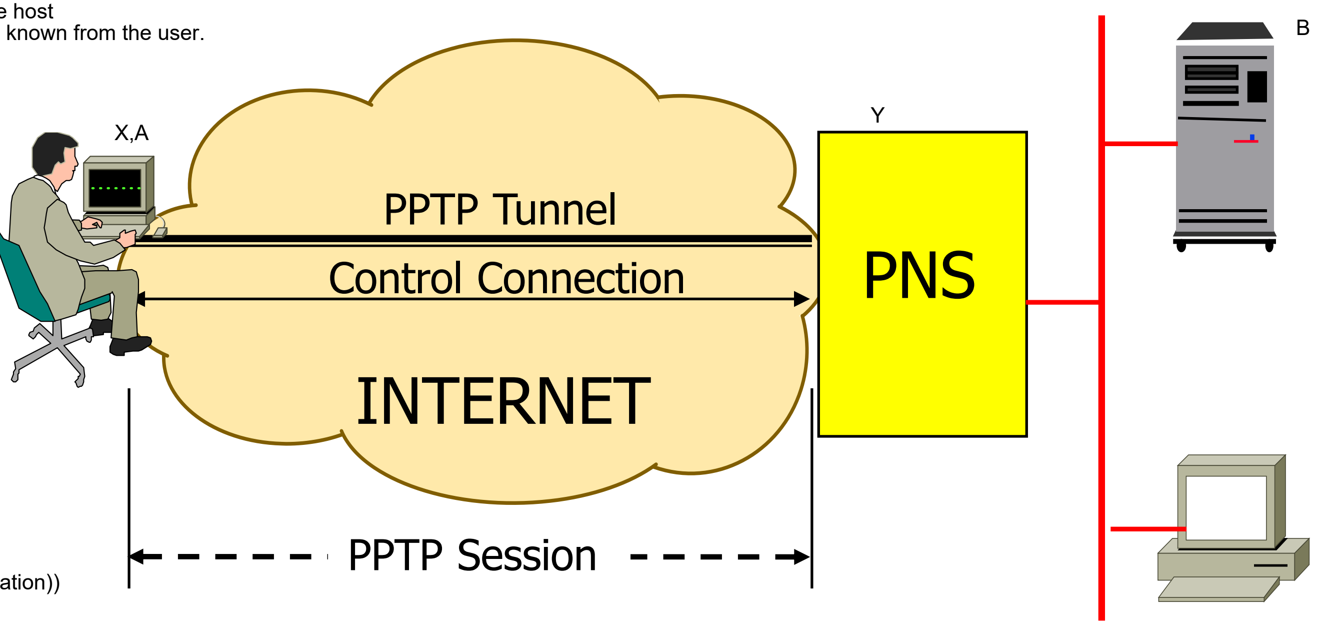
Original Reference Scenario

1) Authenticate urself with PPTP Network Service (PNS):
the host needs an address that allows him to send and receive packets.
the laptop has been configured to know what the ip address of PNS is.
to authenticate he use MS-CHAP, different protocol but same as CHAP.

the host send a request, the server sends a challenge, and the host send the challenge encrypted. But the PNS address has to be known from the user.

MSCHAP/PPTP

Initial Connection: The client wants to connect to the VPN via PPTP and sends a request to the PNS.
Challenge from the PNS: The **PNS** sends a challenge to the client — a unique random number
Client's Response**: The client combines this challenge with the shared password and creates a secret code (hash), which it sends to the **PNS**.
PNS Verification**: The **PNS**, which has the same password, generates the same hash using the challenge it sent and its copy of the password.
If the code matches the client's, authentication is valid, and the connection is established.
* **Security**: As in L2TP, the password is never transmitted in plain text. Only the hash code travels between the client and **PNS**, making the connection more secure.



once the remote host has its own address, give by the corporate network, he can send and receive packets directly from the place he is.

(Ip address of the home network, PNS address | (Assigned Address, Destination))

Customer provisioned deployment mode

General Features

- Adopted by IETF (RFC 2637)
- Tunneling of PPP frames over packet switched networks
- Microsoft Encryption: MPPE
- Microsoft Authentication: MS CHAP
- PPTP Network Server (PNS)
  - Corporate (VPN) gateway
- PPTP Access Concentrator (PAC)
  - For provider provisioned deployment mode

PPP frame

Generic Routing Encapsulation Header(to encapsulate any other protocol packet of every level into an IP packet.)

ip Packet.

PPTP Connections

- **PPTP Data Tunneling**
  - **PPP tunneling**
  - **GRE (of PPP over IP)**

Data-link Header	IP Header	GRE Header	PPP Header	A,B Encrypted PPP Payload (IP Datagram, IPX Datagram, NetBEUI Frame)	Data-link Trailer

x, y

the ppp heaedr was inserted because there was already the implementation to do that, so they left it.

- **Control Connection**
  - **Data tunnel setup, management, and tear-down**
  - **TCP encapsulation**
    - **PNS port 1723**

Data-link Header	IP	TCP	PPTP Control Message	Data-link Trailer

PPTP Header

Length	Message type
Magic cookie	
Data :::	

Value	Control Message
1	**Start-Control-Connection-Request.**
2	**Start-Control-Connection-Reply.**
3	**Stop-Control-Connection-Request.**
4	**Stop-Control-Connection-Reply.**
5	**Echo-Request.**
6	**Echo-Reply.**
7	**Outgoing-Call-Request.**
8	**Outgoing-Call-Reply.**
9	**Incoming-Call-Request.**
10	**Incoming-Call-Reply.**
11	**Incoming-Call-Connected.**
12	**Call-Clear-Request.**
13	**Call-Disconnect-Notify.**
14	**WAN-Error-Notify.**
15	**Set-Link-Info.**

l4 VPN , SSL/ TLS VPNs   to do the tunneling we use ssl or tls(a more strandard version of ssl).

a solution to make the exchange of ip packet secure.

IP sec VPN

cannot go trough nat, because it will change the digest.

VPN based on ipsec, they use it to perform the tunnel.
Usally used into s2s vpn, connect two corporate between a shared network.

to make ip packet secure, ensuring that ip packet are private, have integrity and u can identify the endpoints.
to ensure the integrity of the ip packet, u use cryptography and u need an additional header, called Authentication Header.
We perform with a symmetric or asymmetric key

in the authentication header we have the digest, and other things that tells us what algorith we need to calculate the digest..This defend the IP header to be manomitted.

Authentication Header Protocol (AH)

- Source authentication + data integrity
  - No confidentiality
- AH header inserted between IP header and payload.
  - Protocol field: 51
- Routers process datagrams as usual
  - Not NAT, though

IP header	AH header	data (e.g., TCP, UDP segment)

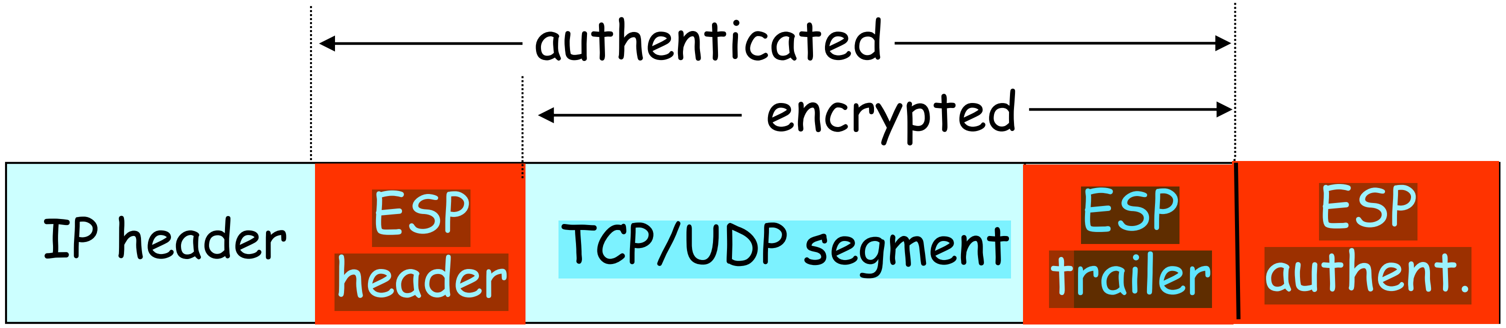
Authentication Header

- SPI: Security Parameter Index
  - Session ID
  - How to verify signature
    - Crypto algorithm
    - Reference to key
- Authentication data
  - Crypto signature
- Next header field
  - Protocol (e.g., TCP, UDP, ICMP) in payload

IP header	AH header	data (e.g., TCP, UDP segment)

Encapsulating Security Payload (ESP)

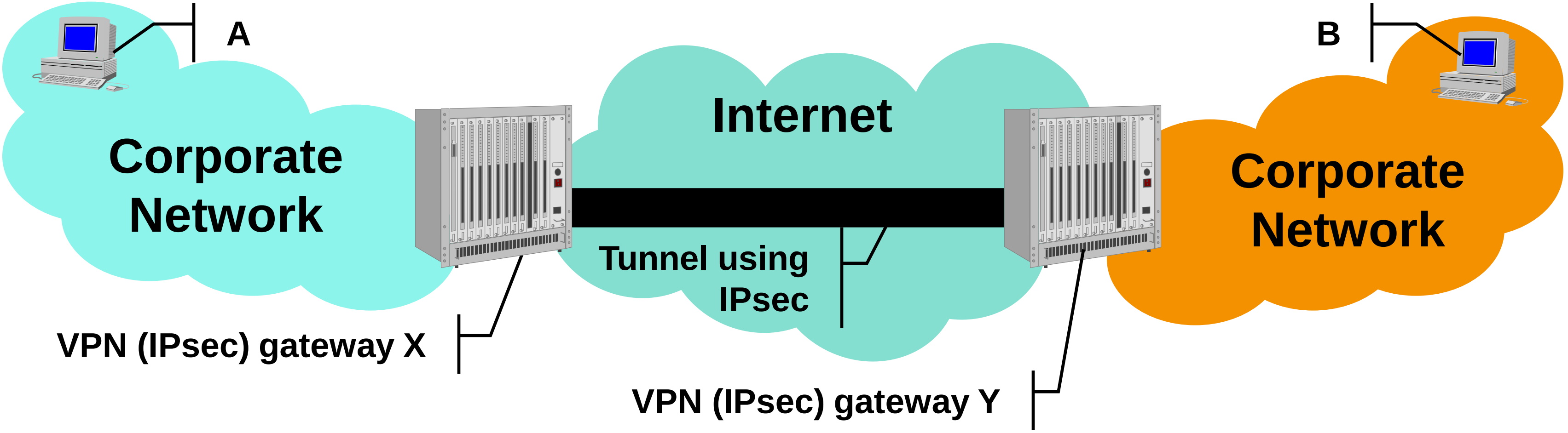
- Data confidentiality
  - Data and ESP trailer encrypted
  - Next header field in ESP trailer
- Host authentication
- Data integrity
  - Authentication field similar to AH
- Protocol = 50



IPsec VPNs

IPsec tunnel between VPN gateways

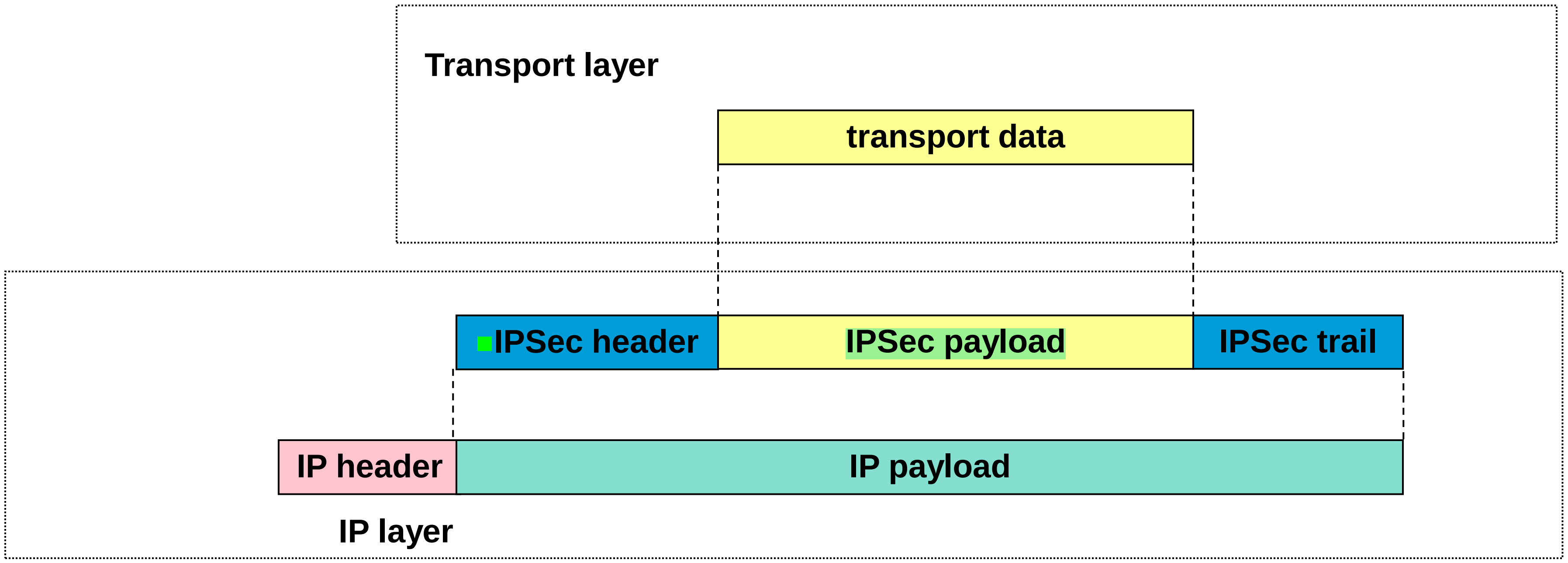
- Encryption
- Authentication
- Encapsulation



Transport Mode

IP header not fully protected

(only authenticated if AH is used)



Tunnel Mode

It protects both IP header and payload